



Spin-momentum locking in a mechanical altermagnet of non-reciprocal actuators

Master's Thesis

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Abstract

Altermagnets are a recently discovered ordered magnetic phase with novel properties promising in spintronics. They are characterized by the rotation relationship between spin sublattices. We study the properties of such system in a classical setting. We build a mechanical apparatus composed of actuators which can be arranged in a variety of geometries, leading to different symmetries. The magnetic spin is replaced by the chirality of oscillators, using non-reciprocal interactions between pairs of actuators. We observe synchronization of actuators and selective propagation in 1D systems. In the altermagnet 2D lattice, we find spin/chirality-momentum locking and a frustrated bistable system. The experimental results are corroborated by theoretical models and simulations.

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1 Introduction

The main purpose of this thesis is to study wave guidance in a mechanical realization of an altermagnet. Wave guidance is the ability to direct waves along a certain path, often with minimal loss or distortion. It is crucial in various applications, from telecommunications to medical imaging. We build a mechanical metamaterial made up of springs and actuators, with the symmetry of an altermagnet. The metamaterial reproduces some properties from the original magnetic phase, with chiral phonons instead of spin up and down electrons.

1.1 Altermagnets

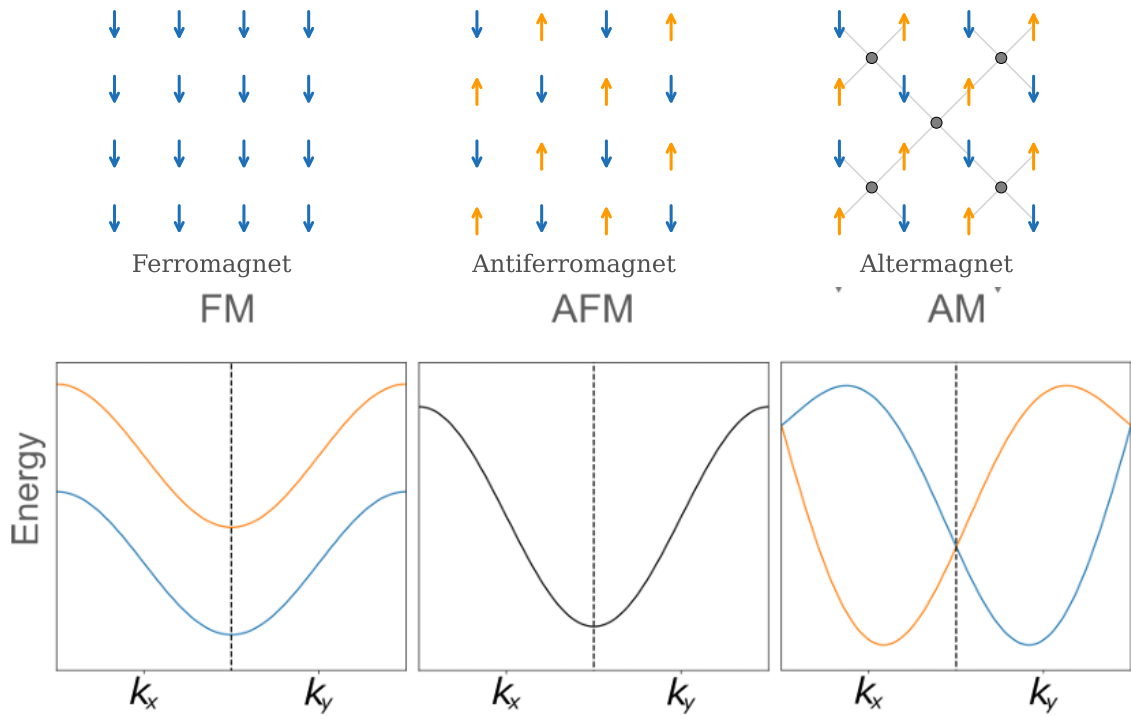


Figure 1: **Collinearly-ordered magnetic phases: ferromagnets, antiferromagnets and altermagnets.** **Top:** diagram of the lattice spins in ferromagnet, antiferromagnet and altermagnet, from left to right. **Bottom:** schematic representation of their band structure. The focus is on the relationship between the spin-up band in orange with the spin-down band in blue.

Before introducing altermagnets, we must briefly mention ordinary collinearly ordered magnetism. In traditional magnets, the magnetic moments of atoms, or spins for short, align in a uniform direction, leading to a net magnetic field. These materials are called ferromagnets. We can also find in nature materials whose spins tend to point in opposite directions to their neighbours, cancelling the net magnetic field. These materials are called antiferromagnets. However, most materials do not exhibit any magnetic order at all. We focus on paramagnets, which are characterized

by a field of randomly oriented spins. The way we classify theoretically different materials in any of these three categories is using symmetries. As we often do, we start with the most symmetric of the three: paramagnets. In these materials, the spins are randomly oriented. The average properties, the expectation values, of the magnetic system are invariant under spin-rotations, which are transformations that only rotate the spin degrees of freedom. As a consequence, paramagnets are also invariant under time reversal, meaning that if we reverse the direction of time, the system looks the same. But what does reversing time mean? It simply means that if we were to film the system with a camera and then play the footage backwards, we wouldn't be able to tell the difference. At the level of equations, what we do is add a minus sign to time. This has an effect on some of the physical quantities. For instance, the sign of the linear momentum changes, because we are playing the movie backwards. The same happens to angular momentum, and more specifically, to spin angular momentum, which also changes sign under time reversal. Time reversal flips a spin up into spin down and vice versa, so it is a particular case of a spin-rotation transformation. Since paramagnets have randomly oriented spins, flipping the spins leaves the system invariant. We say paramagnets are time reversal symmetric. Do antiferromagnets and magnets also have these symmetries? The key is that they do not. Ferromagnets and antiferromagnets break spin-rotation symmetry because the spins align and disalign in a specific direction. Thus, they also break time reversal symmetry. Nevertheless, one can combine spin transformations, acting only on spins, with real space transformations, acting only on the lattice, and obtain symmetries. We consider these two types of transformations to be uncoupled from each other. For antiferromagnets, the combination of time reversal with translation is a symmetry (also with inversion). This happens because translations swap the two opposite spin sublattices. This symmetry along with electrons having half-integer spin (which makes the antiunitary symmetry operator square to -1), leads to Kramer's theorem. The theorem states that every energy level is at least doubly degenerate. In plain language, one can have two electrons, one with spin up and one with spin down, occupying the same energy level, leading to spin degenerate bands. Ferromagnets do not possess this symmetry. Kramer's degeneracy does not hold for them, and they have spin-dependent bands. Spin up and down electrons have different energy band structure and thus they behave differently. So the way we can tell apart different magnetic orders, or phases in general, is by looking at which symmetries they have and which symmetries they break. Now we can tackle altermagnets. Altermagnets were recently discovered because they had strange properties, like anomalous quantum Hall effect. They share some properties with ferromagnets and some other with antiferromagnets. At first they were called anomalous antiferromagnets, but then Smejkal [1] realised they had a different symmetry, so it was a new phase of matter. Altermagnets are another collinearly ordered magnetic phase, but with different symmetries. Like antiferromagnets, they are made up of two spin sublattices, up and down. They break the antiferromagnet symmetry, i.e. the combination of time reversal and translation. However, they are symmetric under the combination of time reversal and a real space rotation. This allows them to have spin-split bands, like ferromagnets. However, altermagnets have no net magnetic field, like antiferromagnets.

We focus on a particular model of altermagnet called the Lieb lattice [3]. The particular sym-

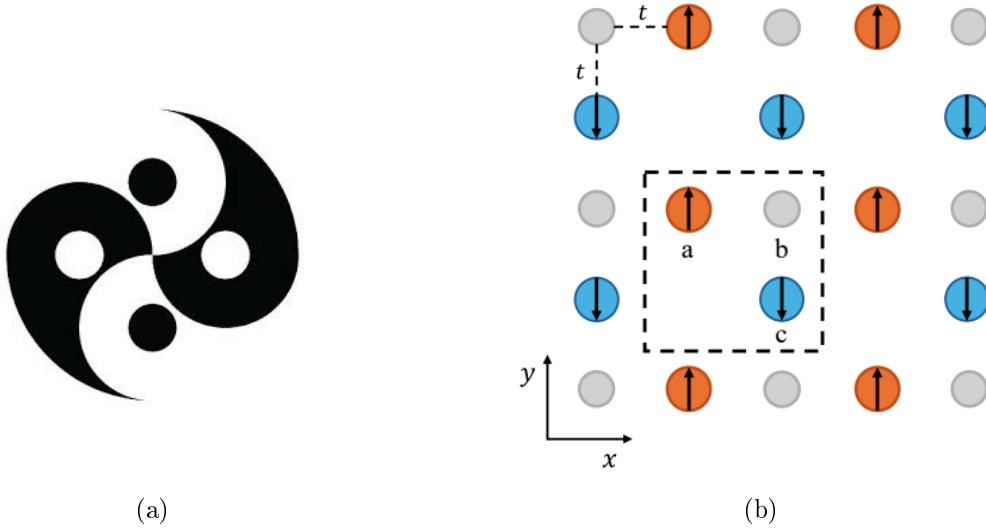


Figure 2: **Systems with $C_4\mathcal{T}$ symmetry.** **2a** Yin-yang symbol with four lobes. **2b** Altermagnetic Lieb lattice [2]. They both share the spin/colour flip combined with 90° rotation symmetry.

metry of this model is time reversal combined with a 90° lattice rotation. We denote it as $C_4\mathcal{T}$. As shown in figure 2, it is a two-dimensional square lattice with a three-atom motif: one spin up atom, one spin down atom, and one neutral atom. The neutral atom is not magnetic, and thus has no spin degree of freedom. On figure 2, one can notice that the nearest neighbours of any spin up site are non-magnetic sites at each side, in the horizontal direction. On the contrary, for spin down sites, the nearest neighbours are non-magnetic sites in the vertical direction. The spin atoms create a magnetic field, which has an effect on the electron spin. The field breaks the $SU(2)$ spin symmetry of electrons down to $\mathbb{Z}_2$. This means that the electron collapses to either spin up or spin down. The energy is minimized when the spin of the electron and atom are aligned. If we only allow hopping between nearest neighbours, we can intuitively predict that spin up electrons are more likely to move in the horizontal direction, while spin down electrons prefer the vertical direction. This means electrons with different spin propagate in different directions. We call this effect spin-momentum locking [4]. The reason why it happens is the symmetry of the altermagnet.

One can generalize the altermagnet symmetry $C_4\mathcal{T}$ to other systems by replacing the spin degree of freedom with other two-level variables. This opens the door to realizing altermagnet-like behaviour in various physical platforms. For example, [5] have replaced spin with the photon's polarization states, and they designed photonic crystals that exhibit altermagnet-like properties. Similarly, [2] used electric dipoles to build an alterelectric. The focus of this thesis is to explore spin-momentum locking in a mechanical Lieb lattice, where the two-level variable is chirality of the oscillator. Spin up and down are replaced by clockwise and counterclockwise motion. We achieve it using non-reciprocity.

1.2 Non-reciprocity

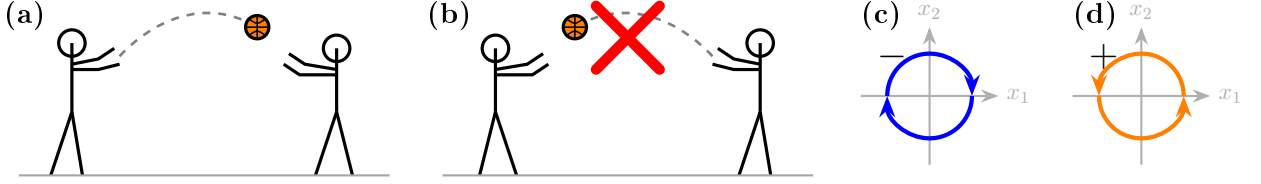


Figure 3: **Schematic representation of non-reciprocity and oscillator chirality.** (a–b) Example of non-reciprocity in basketball. (a) Basketball player A passes the ball to his teammate B. (b) Player B likes shooting the ball himself and does not pass the ball back to player A. (c–d) State space trajectories illustrating the limit cycle dynamics and its chirality. (c) Negative chirality: limit cycle moving in clockwise direction. (d) Positive chirality: limit cycle moving in counterclockwise direction.

To replace the atom’s spin in a quantum altermagnet by an oscillator’s chirality in the mechanical metamaterial, we use non-reciprocity between two linear actuators. Linear actuators are simple springs in which we can program additional interactions. Non-reciprocity is the asymmetry in interactions. The effect of A on B is different from the effect of B on A. Here, A and B are any two subsystems, so this idea occurs in a wide variety of situations. We are interested in coupling two actuators non-reciprocally. To illustrate the idea, we consider a simple example of two springs whose motion only depends linearly on their positions. Newton’s second law is then just $F = Kx$, where F is the force written as a 2-component vector and x is the position displacement from the equilibrium value. Each component of F refers to the force felt by one of the actuators. Then, the stiffness K can be written as a 2×2 matrix, with components

$$K = \begin{pmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{pmatrix}. \quad (1)$$

The diagonal elements of the matrix are just the ordinary onsite linear stiffness values of the springs. The non-diagonal elements couple the position of one of the actuators to the other. Therefore, non-reciprocal means we make the stiffness matrix K asymmetric: $k_{12} \neq k_{21}$. Since we still want the forces F_1 and F_2 to be of same order of magnitude, we often choose a specific non-reciprocal interaction called odd elasticity [6]. The couplings have opposite sign $k_{12} = -k_{21} = k_a$, and thus the stiffness matrix K is antisymmetric. The parameter k_a is called activity or gain, and it will be the only experimental parameter we control and change. The stiffness matrix of such a system of identical oscillators is

$$K = \begin{pmatrix} k_0 & k_a \\ -k_a & k_0 \end{pmatrix}. \quad (2)$$

This choice breaks Newton’s third law of action-reaction and conservation of energy. Therefore, the system becomes active and non-Hermitian. In our experiments we make the stiffness matrix antisymmetric by pumping electrical energy into the actuators.

Odd elasticity allows two actuators to trace out limit cycles in phase space, as we will see in the

following sections. The sign of the activity determines whether the trajectory moves clockwise or counterclockwise. These are the new two-state variables that replace spin. We call them oscillator chirality or mechanical spin. Therefore, two actuators behave as one unit that traces out limit cycles, and that is essentially an oscillator. We can couple several units together in different geometries and with different symmetries. In fact, the thesis is divided into sections according to those geometries. First, we study the behaviour of one unit in section 3. After that, we study two units in section 4, a one-dimensional chain in section 5, the altermagnet cross cell in section 6 and finally the Lieb lattice in section 7. In section 2, we show the experimental setup and a theoretical procedure that maps these classical systems to Hamiltonians. The motivation is to use the intuition and tools from condensed matter theory and tight-binding models. As we mentioned before, energy is not conserved in these oscillator systems. That makes the physics non-Hermitian.

1.3 Non-Hermitian physics

In quantum mechanics, we usually deal with Hermitian observables, which have real eigenvalues and orthogonal eigenvectors. In condensed matter physics, the Hamiltonian operator describes the energy of the system and is typically Hermitian to ensure real energy values. The Hamiltonian also governs the time evolution of quantum states, preserving probabilities. It can have symmetries that lead to conservation laws, which are made explicit by commutation relations. One can then use a simultaneous basis of eigenstates of the Hamiltonian and the symmetry operators. In Hermitian physics there are 3 relevant symmetries: time-reversal symmetry, particle-hole symmetry and chiral symmetry. We have already mentioned the first one. Particle-hole symmetry changes the sign of both energy and charge, swapping particles with holes. Chiral symmetry is a combination of the previous two, and ends up being a sublattice symmetry in many systems. Altland and Zirnbauer used the absence, presence and eigenvalues of these symmetries to classify topological phases of matter the ten-fold way [7].

In many physical systems, especially those that are open or involve gain and loss of energy, particles or information, we encounter non-Hermitian Hamiltonians. The classical systems considered in the project are active, making their Hamiltonians non-Hermitian. There are two main differences with respect to Hermitian physics. First, the energies are complex numbers. The imaginary part determines the gain or loss in probability of the eigenstate in time evolution. If the imaginary part is positive, the state will have gain in probability, and if negative, it will have loss. To analyse the spectrum, we will plot it in the complex energy plane. Note that we added a new dimension to the spectrum compared to the Hermitian case where the energies were located in the real line. This allows for a new type of gap. In Hermitian physics, we had a gap if there was an interval in the real line with no eigenvalues. In non-Hermitian jargon, these gaps are called line gaps. The name comes from the fact that one can draw a line in the complex energy plane to separate eigenvalues. The new type of gap is called a point gap. A point gap exists if there is an eigenvalue in the complex energy plane such that one can draw a closed loop around it that does not cross any other

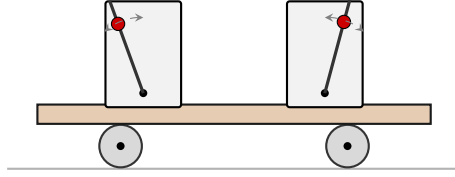


Figure 4: **Synchronization of metronomes.** Illustration of an experiment consisting of two metronomes with different initial conditions and natural frequencies, on top of a board which is supported by cans. After some time, the metronomes synchronize.

eigenvalue. The second difference with respect to Hermitian physics is the extreme sensitivity to boundary conditions. In Hermitian physics, the bulk properties of a system are usually insensitive to the choice of boundary conditions, whether they are periodic or open. It allows to use periodic boundary conditions to simplify calculations without losing generality, and then apply the results to real systems with open boundaries. However, in non-Hermitian systems, this correspondence can break down. The spectrum and eigenstates can change dramatically when switching from periodic to open boundary conditions. For example, a one-dimensional chain with non-reciprocal hopping and periodic boundaries has all its eigenstates extended throughout the chain. If one breaks the chain, so that the boundaries become open, most of the eigenstates will localize at the edges. This model is known as the Hatano-Nelson chain, and the phenomenon is the non-Hermitian skin effect. It is a common theme in Hermitian physics that most eigenstates are extended throughout the bulk system, even for finite systems. There can also be a few edge states localised at the boundaries, as happens for topological insulators. However, in non-Hermitian systems, most eigenstates can become localised at the edges of the system, while only a few remain extended in the bulk. The non-Hermitian skin effect has important implications for the transport properties of non-Hermitian systems.

1.4 Synchronization

We replaced the spins in a magnetic phase by the chirality of oscillators in a metamaterial. As a result, we obtained a set of interacting autonomous oscillators. These systems exhibit collective behaviour like synchronization, wave coarsening [8] and chaos, in different regimes. The most beautiful example of synchronization is placing two (or more) metronomes on a board 4. The board is supported by cans. The metronomes can have different initial conditions and different natural frequencies, and yet after some time they end up synchronizing [9]. The same phenomenon is responsible for fireflies [10], circadian rhythm [11] and many other examples. When oscillators oscillate with the same frequency, but with an arbitrary phase difference, it is called phase locking or synchronization. If the oscillators have a constant phase difference of π or 180° , we refer to it as anti-phase synchronization. And if the oscillators oscillate with the same frequency and same phase, then it is called in-phase synchronization. Along the thesis, we mainly encounter in-phase synchronization. For convenience, we call it simply synchronization. We specify when the phase

difference is nonzero.

There are three requirements for synchronization: the individual units must be autonomous oscillators, they must interact and their individual parameters are similar to each other. The metronomes are autonomous oscillators, they interact through the structure of the board and cans, and their natural frequencies must be sufficiently similar to display synchronization. The non-reciprocal units we introduced before trace out limit cycles and thus behave like two autonomous oscillators. They interact mechanically and weakly (see details in section 2). Again, if the individual parameters are similar, these units will synchronize too.

The dynamics of limit cycles can be separated into two components. There is the amplitude and the phase. There are two types of Lyapunov exponents of the dynamical system. Most of them are negative, and they correspond to the stability of the limit cycle under perturbations in the amplitude. Though at least one of the eigenvalues is zero. It corresponds to phase shifts of the limit cycle trajectory. The fact that the eigenvalue is zero means the phase is neutral under phase perturbations. The phase shifted trajectory is still the limit cycle solution. As a consequence, the global phase of the limit cycle is not important. Since the phase grows with time, we can make time shifts that result in choosing the specific value of the phase desired. For weak interactions between units, the amplitude remains fixed to a constant value. They play no role in the dynamics. The only relevant degrees of freedom then are the phases. This dimensionality reduction in the degrees of freedom is sometimes called the phase dynamics approximation.

We illustrate the simplest example of synchronization. Consider two identical non-reciprocal oscillators coupled together weakly, so that they still follow limit cycles. Therefore, the dynamics is described by the phases of the oscillators. When uncoupled, the oscillators have natural frequencies ω_1 and ω_2 . Since the global phases are unimportant, the interaction terms must depend on the phase difference $\Delta\phi = \phi_2 - \phi_1$. The interaction might be a very complicated function of $\Delta\phi$. Nevertheless, it needs to be 2π -periodic in the phase difference $\Delta\phi$. Therefore, we can expand it as a Fourier series.

$$H(\Delta\phi) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\Delta\phi + b_n \sin n\Delta\phi) \quad (3)$$

There are more constraints on the interaction $H(\Delta\phi)$. The actual mechanism carrying out the interaction in the experiments is a viscoelastic band. The mechanism is reciprocal and satisfies Newton's third law of action and reaction. The interaction that the first oscillator feels is exactly the opposite to the one the second one feels. The interaction terms must be opposite of one another: $H_1(\Delta\phi) = -H_2(\Delta\phi) = H(\Delta\phi)$. They must also conserve the symmetry exchanging the labels of the oscillators $1 \leftrightarrow 2$. This transformation has two effects on the equations. First, it exchanges $H_1 \leftrightarrow H_2$, or more precisely, $H \leftrightarrow -H$. Second, the phase difference changes sign, because we have swapped the order: $\Delta\phi \rightarrow -\Delta\phi$. Since this relabeling should not have any effect on the dynamics, we find the condition $-H(-\Delta\phi) = H(\Delta\phi)$. The interaction must have odd parity to respect the reciprocity between identical oscillators. Therefore, all the even terms in the Fourier series vanish.

We are only left with sine terms. Moreover, as the interaction is weak, we can approximate the interaction term by its leading order term: $\sin \Delta\phi$. Higher order terms oscillate very fast and average out to zero. We arrive at the Kuramoto model [12] of two oscillators:

$$\dot{\phi}_1 = \omega_1 + K \sin(\phi_2 - \phi_1) = \omega_1 + K \sin(\Delta\phi) \quad (4)$$

$$\dot{\phi}_2 = \omega_2 + K \sin(\phi_1 - \phi_2) = \omega_2 - K \sin(\Delta\phi). \quad (5)$$

As the global phase is unimportant, we can focus on the dynamics of the phase difference. The equation is obtained by subtracting (4) and (5). The result is the Adler equation:

$$\frac{d\Delta\phi}{dt} = \Delta\omega - 2K \sin(\Delta\phi). \quad (6)$$

The first term in the RHS is the detuning or mismatch of the natural frequencies. The second term is the leading order contribution of the interactions. Synchronization occurs when the phases are locked together: $\frac{d\Delta\phi}{dt} = 0$. In other words, synchronization solutions are given by the zeros of the RHS of (6). The Adler equation predicts two regimes. If $\Delta\omega < 2K$, the function on the RHS has zeros and the oscillators synchronize. If $\Delta\omega > 2K$, the function has no zeros and the oscillators do not synchronize. The system then exhibits beating, phase drift or chaos.

2 Methods

2.1 Actuators

To replace spin by oscillator chirality, we use two actuators coupled non-reciprocally. We use custom-made voice coil linear actuators, shown in figure 5a. The actuators have several components. They have a passive elastic structure, which is black and orange in figure 5a, that makes it behave like a damped spring. We can also program custom forces into the actuator. The actuator uses the magnetic Lorentz force to displace itself, namely $\vec{F} = L\vec{I} \times \vec{B}$. Therefore, the actuator functions thanks to magnetic field and an electric current. The magnetic field B is generated by a cylindrical iron permanent magnet in the center of the actuator. The current is generated externally with a power supply. The current intensity I goes through a copper coil that surrounds the magnet, which has length L . The magnetic field lines extend outward radially $\vec{B} = B\hat{r}$, while the current winds in the azimuthal direction $\vec{I} = I\hat{\phi}$. As the magnetic field lines cross the coil perpendicularly, the Lorentz force generated is directed along the cylinder: $\vec{F} = LI\hat{\phi} \times B\hat{r} = LIB\hat{z}$. For a given actuator, the factor $k = LB \sin \theta$ is fixed. The only parameter we control is the current. The actuator feels a force linear in the current intensity I going through the coil. Reversing the direction of the current also reverses the direction of the force. The coil is fixed with a 3D-printed passive structure of Tough1500 resin. The force produced moves the ion magnet along the axial coordinate. The spring structure maintains the magnet aligned with the coil. When the magnet is displaced, the spring

structure also compresses.

We are interested in implementing odd elasticity, a force that depends on the position. The actuators use a Texas Instruments linear Hall effect sensor (DRV5057A3) to measure the position of the magnet. As the name suggests, the sensor employs the Hall effect to measure magnetic fields as Hall voltages. Since the magnet is displaced back and forth linearly, the Hall sensor can be used to measure those voltage changes, and transform them into displacements of the position. We can program odd elasticity into the actuators, by connecting them to a custom Printed Circuit Board (PCB) with a simple microprocessor (RP235x). The PCB reads out the Hall sensors at 2 kHz and calculates the current input to the coils at 1 kHz. The firmware is also custom.

2.2 Coupling with neighbours

All the actuators move back and forth in one direction, which we will denote the x direction. We will couple the actuators to their neighbours via a viscoelastic band. The coupling strength is denoted by k_b . Note that the coupling occurs in the directions perpendicular to the motion of actuators, so in the y and z direction. Consider the simplest case in which we have 2 actuators coupled by a viscoelastic band in the y direction. The distance between the two actuators is l . When one of the actuators moves in the x direction, it stretches the viscoelastic band, which creates a restoring force in the x direction. Let $\Delta x = x_2 - x_1$ be the difference in the x positions of the two actuators with respect to their equilibrium position. For a displacement Δx , the viscoelastic band stretches from a distance l to a distance $d = \sqrt{l^2 + \Delta x^2}$. The restoring force acting in the first actuator is given by Hooke's law as $\vec{F} = k_b(\vec{d} - \vec{l})$. But keep in mind that the actuators can only move in the x direction, so they only feel the x component of the force, which is multiplied by $\sin \theta$:

$$F_x = k_b(d - l) \sin \theta = k_b(\sqrt{l^2 + \Delta x^2} - l) \frac{\Delta x}{\sqrt{l^2 + \Delta x^2}} = k_b \Delta x \left(1 - \frac{l}{\sqrt{l^2 + \Delta x^2}} \right) \quad (7)$$

where θ is the angle between the viscoelastic band and the y axis. One would expect that the coupling is linear in Δx due to Hooke's law. However, the geometry of the coupling, in which the actuator displacement and the viscoelastic band are not collinear, leads to the appearance of a nonlinear factor in the coupling. This is known as geometric stiffening.

We can consider the limiting cases of equation (7). If the x displacements are very large in comparison with l , then the nonlinear factor vanishes and the force becomes linear in the displacement. If the x displacements are small with respect to l , then we can Taylor expand the nonlinear factor as

$$\left[1 + \left(\frac{\Delta x}{l} \right)^2 \right]^{-1/2} \approx 1 - \frac{1}{2} \left(\frac{\Delta x}{l} \right)^2 + \mathcal{O} \left(\left(\frac{\Delta x}{l} \right)^4 \right). \quad (8)$$

In that case, the force acting on the actuator in the x direction is approximately given by

$$F_x \simeq k_b \frac{\Delta x^3}{l^2}, \text{ for } \Delta x \ll l. \quad (9)$$

In our setup, the amplitude of each actuator is at most 3 mm. As a consequence, $\Delta x < 6$ mm. The smallest distance in y direction between actuators is $l = 35$ mm. This means we are in the regime of small displacements, and we need to use equation (9). Note that this equation is not linear in Δx , but it is cubic. We expect the coupling in the direction perpendicular to displacement to be smaller than in the collinear case. It makes sense because the cubic term is smaller than the linear term, and so the coupling is weaker than ordinary Hooke's law. An additional thing to note is that we can recover the linear coupling by pre-tensioning the viscoelastic band. In the experimental setups, we use the band to keep the actuators straight and keep them from being pulled down by gravity. So all the actuators move the band in the same way, and the band remains unstretched. To conclude, the coupling between neighbour actuators is given by formula:

$$F = \frac{k_b}{2l^2}(x_2 - x_1)^3. \quad (10)$$

2.3 Experimental setup

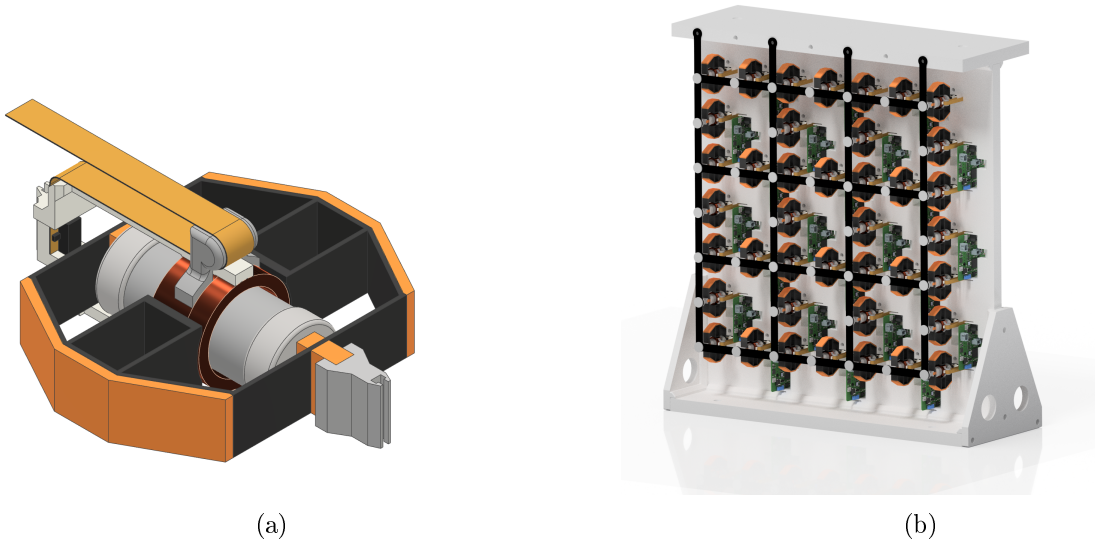


Figure 5: **Actuator and experimental apparatus.** **5a** Voice coil linear actuator with a cylindrical iron magnet, a copper coil that winds around it, and a elastic structure that keeps the former and the latter aligned. The actuator operates due to the magnetic Lorentz force. A current is provided through the coil (angular), which interacts with the radial magnetic field to yield a displacement in the axial (z) direction. **5b** Mechanical Lieb lattice made out of coupled actuators. The apparatus has two layers of actuators arranged in a Lieb lattice. The two layers are separated by a wall, to which the actuators are fixed. The interaction within the layers is through viscoelastic bands. The interaction between the layers is odd elasticity, only between mirror actuators.

The experimental setup is made up of two 2D layers of actuators. The two layers are separated by a wall, where we attach the actuators. There are two types of interactions between the actuators. First, there is the odd elasticity we already introduced in 1.2. This interaction is programmed into the PCB connecting the two actuators. Each actuator is placed to one side of the wall. Altogether they form a non-reciprocal unit. The second type of interaction consists of a viscoelastic band attached to the ends of the actuators, which we introduced in section 2.2. The band is fixed at the top of the wall structure to counteract the effect of gravity. However, it is fixed neither at the bottom nor on the sides.

The actuators making up the layers on either side of the wall interact through the viscoelastic band. The interaction between actuators on different sides of the wall is exclusively odd elasticity. The setup can also be interpreted as one 2D layer of non-reciprocal units. We can arrange the actuators in different configurations. This allows the study of systems with different symmetries and dimensions. Some examples are 1D chains, cross cells and the altermagnet $C_4\mathcal{T}$ symmetry of the Lieb lattice. Besides the way we place the actuators and couple them, the only parameter we control is the activity k_a of the actuators.

In order to carry out measurements, we use a set of high-speed cameras called Qualisys Arqus 5 to track the position of the actuators. The cameras record the position in 3D space, sampling at a rate of $f_s = 500$ Hz. We sometimes calculate the oscillation frequency by performing a Fourier transform. The precision of the frequency results depends on the recording time of the experiment, assuming the system is in the steady state. As the frequency resolution of the Fourier transform is $\Delta f = \frac{f_s}{N} = \frac{f_s}{f_s T} = \frac{1}{T}$. In the previous formula, $N = f_s T$ is the total number of samples taken and T is the total time of recording. Therefore, we increase the precision of the measurements by recording for longer times.

2.4 General theoretical procedure

As we have mentioned, we want to study systems made of classical springs with different geometries. In this section we outline the steps involved in the theoretical description of a general system.

The first step is to write down the equations of motion for the system. To do so, we use Newton's second law and add all relevant forces acting on the system. The forces we consider are: linear elastic stiffness of the spring, damping force due to the spring's friction, nonlinear saturation of the spring, coupling to neighbours via a viscoelastic band and coupling to neighbours via odd elastic interactions. All the force terms except the latter can be realized in experiments mechanically. For the latter, we need to plug energy into the system. If N is the number of springs of our system, we will have a system of N coupled second-order ordinary differential equations. Note that the number of degrees of freedom is $2N$, because we have the position and velocity of each spring.

The second step is to perform an asymptotic expansion of the solution. Solving the whole system is very difficult. One way to simplify the problem is to identify which force terms are the strongest and which are the smallest. We can treat the smaller terms as perturbations to the strongest one. This divides the problem into solving the unperturbed system, without the small perturbations, and then adding the effect of the perturbations. We denote the smaller terms by adding a factor of $\epsilon \ll 1$ in front of them. Intuitively, the strongest term will have a much faster timescale than the smaller ones. The springs will exhibit several of these timescales in their periodic motion. The asymptotic expansion allows us to study the different timescales separately. This can be done in practice in two ways. The first is the method of averaging. This method removes the fast timescale by averaging over its period. Then, the resulting equations describe the slow dynamics, and one can then solve them. The second way is the multiple scale expansion, which is more rigorous but uses the same idea. In the next section 2.5, we show the steps of the method for a simple example. The method introduces different time variables, each denoting one of the timescales. We restrict ourselves to two timescales. The problem is then divided into two equations with different timescale magnitudes, and the ODE has transformed into a PDE. Then, one can solve the fast timescale equation (order ϵ^0). Notice that the two constants of integration from solving the partial differential equation now become coefficients depending on the slow time variable. These coefficients are complex if one considers as basis a Fourier series: $e^{in\omega t}$. It is also important to note that the couplings in the classical equations of motion are real, which constrains the two coefficients to be complex conjugates of each other. One then uses that solution to solve the slow timescale equation (order ϵ). The slow timescale equation contains secular terms, which are terms that grow unboundedly. They arise from the asymptotic expansion and from the fact that the fast timescale solution contains free parameters to be determined. We can choose the free parameters such that the secular terms vanish. This is called the solvability condition, and it comes from the Fredholm alternative theorem in analysis. It states that a linear inhomogeneous equation has a solution if and only if the inhomogeneous term is orthogonal to the null space of the adjoint operator of the differential equation. The solvability condition leads to new equations that govern the slow time dynamics of the free parameters. These equations are called amplitude equations, modulation equations, envelope equations or Stuart-Landau equations. So we have rewritten the original degrees of freedom $(x, \dot{x})$ in terms of a complex valued coefficient we denote by $A(T)$. And we have replaced the classical equations of motion by the amplitude equations for $A(T)$. The important difference is that $A(T)$ only contains the slow timescale dynamics, while the fast timescale dynamics is encoded in the basis functions $e^{in\omega t}$. Moreover, the amplitude equations are first order ODEs, while the original equations of motion were second order ODEs. The reason for this is the relation between the position and velocity in an oscillatory system, which is a factor $i\omega$. Therefore, the linearized dynamics are described by an evolution matrix analogous to the Hamiltonian of the Schrödinger equation. Studying its spectrum and eigenvectors gives us useful information about the system. We actually compared these evolution matrices with tight-binding quantum Hamiltonians, to get intuition. In the next section, we will see how to apply the multiple scale expansion to a simple example, the weakly damped harmonic oscillator. Note that ω is just a parameter that we can choose, and it is usually chosen to be the natural frequency of the unperturbed system.

The third step is to replace the complex modulation coefficients by their amplitude/radius and phase: $A(T) = R(T)e^{i\theta(T)}$. This step again separates the dynamics into 2 equations, one for the radius and one for the phase. Thus there is one radius and one phase equation for each spring.

The next step is to reduce the number of equations. We can do so by noting that the dynamics of the springs are stable limit cycles (in general). They are characterized by the fast decay of perturbations in the amplitude/radius of oscillation. In rigorous terms this means that the Lyapunov exponents of the amplitude components are negative, and perturbations decay exponentially fast. However, limit cycles have a zero mode, which corresponds to phase changes. Perturbations in the phase neither grow nor decay. The Lyapunov exponent associated to the phase is zero. Therefore, deviations in the radius/amplitude are much smaller than deviations in the phase. One can then take the phase dynamics approximation, which consists in assuming that the radius/amplitude of the limit cycle is (periodically) constant. In other words, the shape of the limit cycle is fixed. One ends up with a system of N coupled first order ODEs for the phase of each spring.

One can also carry out the procedure for a system with periodic boundary conditions. With these boundaries it is easier to study the infinite size limit. The way to go is to do a Fourier transformation to momentum space. The difference is one is now working with the momentum basis, and the displacements are not labelled by position, but by momentum. Then one can follow the same steps as before.

2.5 Multiple scale expansion

The method of multiple scales is a powerful technique for analyzing nonlinear ordinary differential equations that exhibit dynamics on widely separated timescales. To illustrate the approach, consider the weakly damped harmonic oscillator described by the equation

$$\ddot{x} + x + 2\varepsilon\dot{x} = 0,$$

where $\varepsilon \ll 1$ represents a small dimensionless parameter quantifying the weak damping. A naive perturbation approach would seek an approximate solution as a power series expansion in ε :

$$x(t) = x_0(t) + \varepsilon x_1(t) + \mathcal{O}(\varepsilon^2).$$

Substituting this ansatz into the governing equation yields the hierarchy of equations

$$[\ddot{x}_0 + x_0] + \varepsilon[\ddot{x}_1 + x_1 + 2\dot{x}_0] + \mathcal{O}(\varepsilon^2) = 0.$$

Solving these equations order by order gives the approximate solution

$$x(t, \varepsilon) = \sin t - \varepsilon t \sin t + \mathcal{O}(\varepsilon^2).$$

However, this approximation becomes invalid for times $t > \mathcal{O}(1/\varepsilon)$ due to the presence of secular terms that grow without bound. To obtain an approximation valid on longer timescales, we introduce two independent time variables: the fast time $\tau = t$ and the slow time $T = \varepsilon t$. The solution is then sought as a function of both timescales:

$$x(t, \varepsilon) = x_0(\tau, T) + \varepsilon x_1(\tau, T) + \mathcal{O}(\varepsilon^2).$$

The time derivative transforms according to the chain rule:

$$\frac{d}{dt} = \partial_\tau + \varepsilon \partial_T,$$

so that

$$\dot{x} = \partial_\tau x_0 + \varepsilon(\partial_T x_0 + \partial_\tau x_1) + \mathcal{O}(\varepsilon^2).$$

At leading order ($\mathcal{O}(\varepsilon^0)$), we obtain

$$\partial_{\tau\tau} x_0 + x_0 = 0,$$

which admits the general solution

$$x_0(\tau, T) = A(T) \sin \tau + B(T) \cos \tau,$$

where the coefficients A and B are allowed to vary slowly with T . At next order ($\mathcal{O}(\varepsilon)$), we find

$$\partial_{\tau\tau} x_1 + x_1 = -2(\partial_T x_0 + \partial_\tau x_0). \tag{11}$$

To avoid secular terms that would render the perturbation expansion invalid for $t \sim 1/\varepsilon$, we require the right-hand side to be equal to zero. This is called the solvability condition. The underlying reason behind it is a mathematical theorem called the Fredholm alternative. It states that there are two possibilities or alternatives for a general operator. If the homogeneous equation of the aforementioned operator only has the trivial solution, it means that the operator is nonsingular and the inhomogeneous equation will have a unique solution for any source term. The alternative is that if the homogeneous equation has nontrivial solutions, then the operator is singular, and there is no guarantee of solutions for a general source term. In the former case, the nullspace of the operator is trivial, it contains only the zero vector. The dimensionality of the nullspace and range of an operator are related and have to sum to the dimension of the vector space. As a consequence, the range of the operator in the first case is the whole vector space, and that includes any source term. If the nullspace has nonzero dimensionality, that means that the range doesn't include the whole vector space. In other words, there are some source terms for which the inhomogeneous equation has no solution. Mathematically, those terms form the nullspace of the adjoint operator. We can call

them homogeneous adjoint solutions. If we require the equation to have a solution, we must avoid them. The projection of the source term with the homogeneous adjoint modes must be set to zero. In physics terms, the nontrivial homogeneous solutions are plane waves with the frequency of the operator. That is precisely the resonant frequency of the system. If one adds a resonant-frequency source term then the system blows up. That is precisely what happens in equation (11). In the example, the operator is the harmonic oscillator and x_0 is the solution to the homogeneous equation, i.e. the equation at order $\mathcal{O}(\varepsilon^0)$. The equation at order $\mathcal{O}(\varepsilon)$ is again the same harmonic oscillator for x_1 , with some additional source terms linear in x_0 . Thus, the source terms oscillate with the resonant frequency, and they are called secular. For the solution to be well-defined, periodic and bounded, the secular terms must cancel out, yielding the solvability condition.

This solvability condition yields the differential equations

$$A' + A = 0 \quad \text{and} \quad B' + B = 0,$$

where primes denote differentiation with respect to T . These equations are called modulation equations, as they govern the slow evolution of the oscillation amplitude and phase. They are first order coupled linear ordinary differential equations that can be solved straightforwardly. However, we will gain more insight by writing them as Schrödinger-like equations. We can do so by writing the equations in matrix form, and multiplying by i :

$$i \begin{pmatrix} A' \\ B' \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \quad (12)$$

The matrix on the right-hand side can be interpreted as an effective non-Hermitian Hamiltonian describing the slow dynamics of the amplitudes A and B . All the system's properties are encoded in this matrix. Therefore, in general we will compute the eigenvalues and eigenvectors of such effective Hamiltonians and interpret them to understand the long-time behaviour of the original system. The solution can be written as a combination of the eigenmodes of the Hamiltonian, each evolving in time with a phase according to its eigenvalue e^{-iEt} .

Back to the example, the Hamiltonian in this case is $H = -iI$, where I is the identity matrix. Then, the eigenvalues are both $-i$, which have negative imaginary part. This means that both amplitudes decay exponentially in time. The solutions are $A(T) = A_0 e^{-T}$ and $B(T) = B_0 e^{-T}$. Substituting back, we obtain the uniformly valid approximation

$$x(t, \varepsilon) = e^{-\varepsilon t} (A_0 \sin t + B_0 \cos t) + \mathcal{O}(\varepsilon),$$

which correctly captures the slow exponential decay of the oscillation amplitude. This example demonstrates the essential ideas of the multiple scales method, which can be extended to more complex nonlinear systems exhibiting multiple timescales.

3 One non-reciprocal unit

In this section we implement mechanical spin or oscillator chirality using two linear actuators interacting non-reciprocally. We show the theoretical modelling first and afterwards the experimental results.

3.1 Theory

We consider the simplest model possible to replace spin: two linear actuators coupled non-reciprocally. We denote the two degrees of freedom as x and y , which correspond to the displacement of the actuators from the equilibrium position. We choose the specific non-reciprocal interaction to be odd elasticity. One can interpret the system as a 2D oscillator. The equations of motion are

$$\ddot{x} + k_0 x + \gamma \dot{x} + k_c x^3 + k_a y = 0 \quad (13)$$

$$\ddot{y} + k_0 y + \gamma \dot{y} + k_c y^3 - k_a x = 0 \quad (14)$$

where k_0 is the harmonic linear stiffness, γ is the damping coefficient, k_c is a nonlinear saturation coefficient that stabilizes the dynamics, and k_a is the non-reciprocal coupling strength, also called the activity or gain. There are different ways to analyse the system. One way is to plug in the plane wave ansatz $x(t) = X e^{-iEt}$ and $y(t) = Y e^{-iEt}$ in the linearized equations, and solve for the frequencies/energies E . Another method is to perform the multiple scale expansion, as explained in the section 2.5. The energies of the linearized system are

$$\text{clockwise: } E_1 = -\frac{i}{2} \left(\gamma + \frac{k_a}{\sqrt{k}} \right), \quad (15)$$

$$\text{counterclockwise: } E_2 = \frac{i}{2} \left(-\gamma + \frac{k_a}{\sqrt{k}} \right), \quad (16)$$

and the eigenvectors are

$$\text{clockwise } v_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (17)$$

$$\text{counterclockwise } v_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}. \quad (18)$$

On the one hand, E_1 has 2 eigenvectors in which the oscillator rotates clockwise in the plane. What distinguishes these eigenvectors is their initial phase, basically a global phase. As the time evolution is given by the phase factor e^{-iEt} , these modes are exponentially suppressed as time evolves. On the other hand, E_2 has 2 eigenvectors that rotate counterclockwise, each with a different initial phase. Note that the energies are complex as a consequence of the absence of energy conservation. There are two regimes for the system described by equations (13) and (14). If $|k_a| < \gamma\sqrt{k}$, so for low values

of activity, the energies have all a negative imaginary part. This means that the oscillations of the actuators are exponentially damped, because friction dominates. The origin is a stable fixed point, so the system becomes at rest and there is no motion. However, if the activity $|k_a|$ is increased, the system undergoes a supercritical Hopf bifurcation. For $|k_a| > \gamma\sqrt{k}$, the origin is unstable and a stable limit cycle appears. The cubic term $k_c x^3$ is necessary for the limit cycle to be stable. The sign of the activity k_a determines the direction of the limit cycle. If $k_a > 0$, the limit cycle has counterclockwise direction, and if $k_a < 0$, it has clockwise direction. This feature can be seen in equations (15) and (16) by changing the sign of k_a . If the sign is changed, the modes that can have the limit cycles, only for sufficiently large activity, are the ones corresponding to the clockwise chirality.

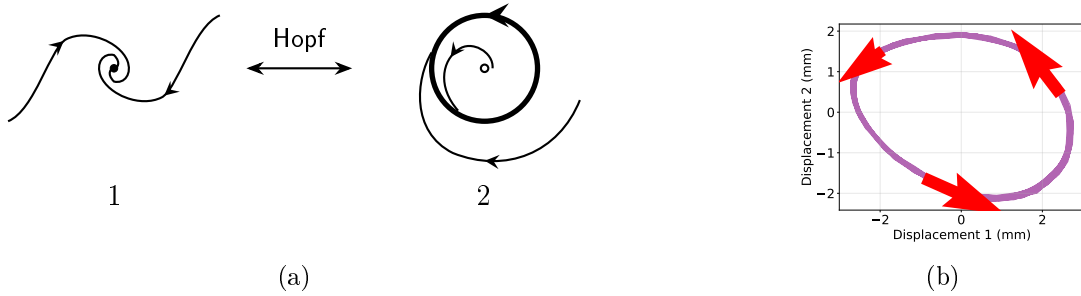


Figure 6: **Dynamics of one unit.** **6a** Regimes of the dynamics separated by a supercritical Hopf bifurcation. **6a1.** Overdamped regime: $|k_a| < \gamma\sqrt{k_0}$, oscillations die. **6a2.** Oscillations regime: $|k_a| > \gamma\sqrt{k_0}$, stable limit cycles with chirality: positive ($k_a > 0$ counterclockwise) or negative ($k_a < 0$ clockwise). **6b** Experimental data in state space displaying the limit cycle.

By plotting the trajectory of the system in state space, i.e. in the x - y plane, we can see the effect of the non-reciprocal coupling. The system traces out limit cycles above the Hopf bifurcation. Limit cycles are isolated closed trajectories. The fact that they are closed means that they are periodic solutions. They are usually defined in phase space, and they are best visualized when the phase space is two-dimensional. However, our system of equations (13) and (14) has actually four degrees of freedom: the displacements x and y , and their velocities $\dot{x}$ and $\dot{y}$. The phase space is thus four-dimensional. However, we can project the trajectories onto the x - y plane, and we will see that they also trace out limit cycles there. Systems with negative chirality trace out limit cycles in clockwise direction, in which the oscillations of spring x go behind in phase with respect to those of spring y . Meanwhile systems with positive chirality trace out limit cycles in counterclockwise direction, making spring x go ahead in phase with respect to spring y . This system's limit cycles are very specific. From the eigenvector formulas (17) and (18), we can observe that the two linear actuators differ in value by a constant factor $\pm i = e^{\pm i\pi/2}$. It means that the two actuators have a phase difference of $\pm\pi/2$, and they are said to be in quadrature.

In conclusion, we use odd elasticity (or non-reciprocity in general) between two linear actuators to achieve stable limit cycles that can oscillate either in clockwise or counterclockwise direction, and they can be used to replace the spin. We refer to the mechanical analogue of spin as oscillator chirality or mechanical spin.

3.2 Hopf bifurcation

Consider two actuators with the following interactions, which we consider to be the most important forces acting on the system: linear stiffness, friction, odd elasticity and cubic stiffness. The classical equations of motion are (13) and (14). We can study the linearized system at the origin. The Jacobian at that point is

$$J(0,0,0,0) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k_0 & k_a & -\gamma & 0 \\ -k_a & -k_0 & 0 & -\gamma \end{pmatrix} \quad (19)$$

and the eigenvalues are

$$\lambda = \frac{1}{2} \left(-\gamma \pm \sqrt{\pm 4ik_a - 4k_0 + \gamma^2} \right) \quad (20)$$

If we substitute the critical value $k_a^c = \gamma\sqrt{k_0}$, then one can complete the square inside the square root, and two of the eigenvalues become purely imaginary: $\lambda = \pm i\sqrt{k_0}$. The fact that two complex conjugate eigenvalues have vanishing real part is the defining property of a Hopf bifurcation. To study the details of the bifurcation, we can perform a change of variables that reduces the order of the ODEs:

$$x(t) = A(t)e^{i\omega_1 t} + A^*(t)e^{-i\omega_1 t} = z + z^* \quad (21)$$

$$\dot{x}(t) = i\omega_1(A(t)e^{i\omega_1 t} - A^*(t)e^{-i\omega_1 t}) = i\omega_1(z - z^*) \quad (22)$$

$$z = x - \frac{i}{\omega_1} \dot{x} \quad (23)$$

The new variable z depends on the position and velocity of the actuator. We carry out the same change of variables for y , and we denote the new variable by f . Substituting in the equations of motion, we find that the equations in the new variables are

$$\dot{z} = (-\gamma + i\omega_1)z - \left(\alpha + \frac{3ik_c}{\omega_1} \right) |z|^2 z - \frac{ik_a}{\omega_1} f, \quad (24)$$

$$\dot{f} = (-\gamma + i\omega_1)f - \left(\alpha + \frac{3ik_c}{\omega_2} \right) |f|^2 f + \frac{ik_a}{\omega_2} z. \quad (25)$$

These are Stuart-Landau equations [13, 14, 15]. It is also the normal form of a Hopf bifurcation. One can obtain the usual normal form in polar coordinates by substituting $z = Re^{i\theta}$. One finds

$$\dot{R}_1 = -\gamma R_1 + \frac{k_a}{\omega_1} R_2 \sin \Delta\phi, \quad (26)$$

$$\dot{R}_2 = -\gamma R_2 + \frac{k_a}{\omega_2} R_1 \sin \Delta\phi, \quad (27)$$

$$\dot{\phi}_1 = \omega_1 - \frac{3k_c}{\omega_1} R_1^2 - \frac{k_a}{\omega_1} \frac{R_2}{R_1} \cos(\phi_2 - \phi_1), \quad (28)$$

$$\dot{\phi}_2 = \omega_2 - \frac{3k_c}{\omega_2} R_2^2 + \frac{k_a}{\omega_2} \frac{R_1}{R_2} \cos(\phi_2 - \phi_1), \quad (29)$$

$$\dot{\Delta\phi} = \omega_2 - \omega_1 - 3k_c \left(\frac{R_2^2}{\omega_2} - \frac{R_1^2}{\omega_1} \right) + k_a \cos \Delta\phi \left(\frac{R_1}{R_2 \omega_2} + \frac{R_2}{R_1 \omega_1} \right). \quad (30)$$

Looking at the radial equations, the activity k_a is the responsible for the instability at the origin, while the friction γ stabilizes the same point. It is the same type of equations as the normal form of a supercritical Hopf bifurcation. The difference with the archetypical equations is that here the instability is driven by odd elasticity instead of negative damping.

We can calculate the critical value of activity at the Hopf bifurcation by using a plane wave ansatz. We start by defining a different complex variable $z = x_1 + ix_2$. Close to the Hopf bifurcation, we can neglect the nonlinear terms and the equations of motion reduce to one complex second order ODE:

$$\ddot{z} + k_0 z + \gamma \dot{z} - i k_a z = 0. \quad (31)$$

It is essentially a competition between damping γ and active coupling k_a . Assume a solution of the form $z = e^{\lambda t}$:

$$\lambda^2 + \gamma \lambda + k_0 - i k_a = 0. \quad (32)$$

A Hopf bifurcation happens precisely when $\text{Re}(\lambda) = 0$ and $\text{Im}(\lambda) \neq 0$, we can substitute $\lambda = i\omega_c$ into the characteristic equation:

$$-\omega_c^2 + i\gamma\omega_c + k_0 - i k_a = 0. \quad (33)$$

Solving for the real and imaginary parts independently

$$\text{Real part: } -\omega_c^2 + k_0 = 0 \implies \omega_c = \sqrt{k_0}, \quad (34)$$

$$\text{Imaginary part: } \gamma\omega_c - k_a = 0 \implies \omega_c = \frac{k_a}{\gamma}, \quad (35)$$

we find again the critical coupling threshold k_a^c :

$$k_a^c = \gamma \sqrt{k_0}. \quad (36)$$

To know the behaviour of the limit cycle close to the bifurcation, we perform stability analysis.

We substitute $z = Re^{i\omega t}$ for radius/phase separation, and by setting $\dot{R} = 0$ we find:

$$-\omega^2 R + k_0 R + i\omega\gamma R + k_c R^3 - ik_a R = 0. \quad (37)$$

We can separate again into real and imaginary parts:

$$\text{Real part: } -\omega^2 + k_0 - k_c R^2 = 0 \implies \omega^2 = k_0 + k_c R^2, \quad (38)$$

$$\text{Imaginary part: } \omega\gamma = k_a \implies \omega = \frac{k_a}{\gamma}. \quad (39)$$

The imaginary equation indicates that the frequency of the limit cycle scales linearly with the activity. Combining both parts allows solving for the radius R^2 :

$$\frac{k_a^2}{\gamma^2} = k_0 + k_c R^2 \quad (40)$$

$$R^2 = \frac{1}{k_c} \left(\frac{k_a^2}{\gamma^2} - k_0 \right), \quad (41)$$

$$R^2 = \frac{1}{k_c \gamma^2} (k_a^2 - \gamma^2 k_0). \quad (42)$$

We use the previous result $(k_a^c)^2 = \gamma^2 k_0$ to simplify

$$R^2 = \frac{1}{k_c \gamma^2} (k_a^2 - (k_a^c)^2) = \frac{1}{k_c \gamma^2} (k_a - k_a^c)(k_a + k_a^c), \quad (43)$$

and approximating near the threshold where $k_a + k_a^c \approx 2k_a^c$, we obtain

$$R \approx \sqrt{\frac{2k_a^c}{k_c \gamma^2} (k_a - k_a^c)}. \quad (44)$$

This confirms the square-root behavior for the limit cycle radius above the threshold, together with the linear behaviour of the frequency.

Far from the Hopf bifurcation, one needs to include higher order terms $A^5, A^7 \dots$ that give exponents $1/4, 1/6 \dots$ to the activity for the function of the radius. These higher order terms make the slope flatter until it is constant.

The limit cycles we discussed occur in the 4-dimensional phase space $(x_1, \dot{x}_1, x_2, \dot{x}_2)$. Nevertheless, we can project the trajectory to the 2-dimensional state space (x_1, x_2) . As the phase difference between the two springs is enforced to be $\Delta\phi = \frac{\pi}{2}$ by the odd elastic interactions, the trajectory in the lower-dimensional space is an ellipse.

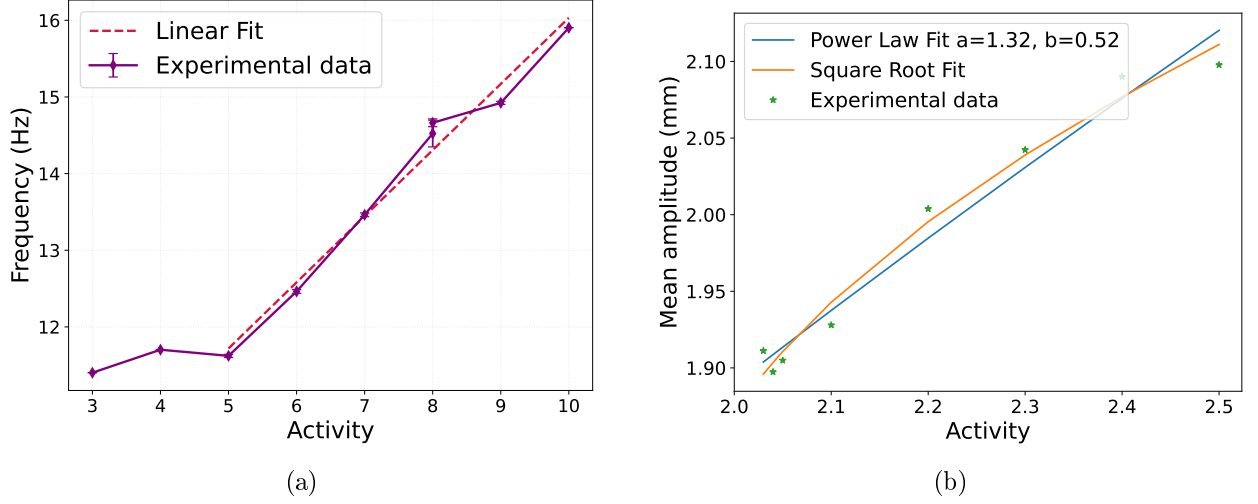


Figure 7: **Limit cycle scalings close to Hopf bifurcation.** **7a** Linear scaling of frequency with activity parameter. **7a** Square root scaling of amplitude with activity parameter. The experimental results agree with the theoretical predictions of the supercritical Hopf bifurcation.

3.3 Experiments

The regime below the Hopf bifurcation, where the system goes to rest, is not interesting, so we focus on the regime above the Hopf bifurcation where the system traces out limit cycles. There is only one parameter to control in the experiments, and that is the activity k_a . The rest of coupling coefficients in equations (13) and (14) are determined by the actuators' mechanical properties, which we don't change. We can decompose the analysis of the limit cycles into two: the phase and the amplitude.

We observe that the frequency of the limit cycle increases when we increase the gain, see figure 8a. Moreover, the frequency increases linearly, as predicted by the supercritical Hopf bifurcation. The measurements of the amplitude of oscillation have great variability. Both actuators forming the unit oscillate with the same frequency, because they are coupled non-reciprocally. But they don't oscillate with quite the same amplitude, and this is due to small differences in each of the actuators. Keep in mind that the actuators are several years old because they were made for another setup. Some of them were crooked or broken and had to be repaired, and that led to the system being considerably inhomogeneous. For that reason, we also measured the motion of another unit. The amplitude and frequency of the limit cycle are shown in figure 8b. Again, we see that the frequency increases linearly with activity. At lower values of activity, the slope of the frequency flattens. The amplitude stays roughly the same for a big range of activities. This observation might be a consequence of the actuators having a hard stop at their ends. Above a certain value of activity, the actuator reaches its maximum displacements. Increasing the activity beyond doesn't make the amplitude grow further, but it does make the oscillations go faster.

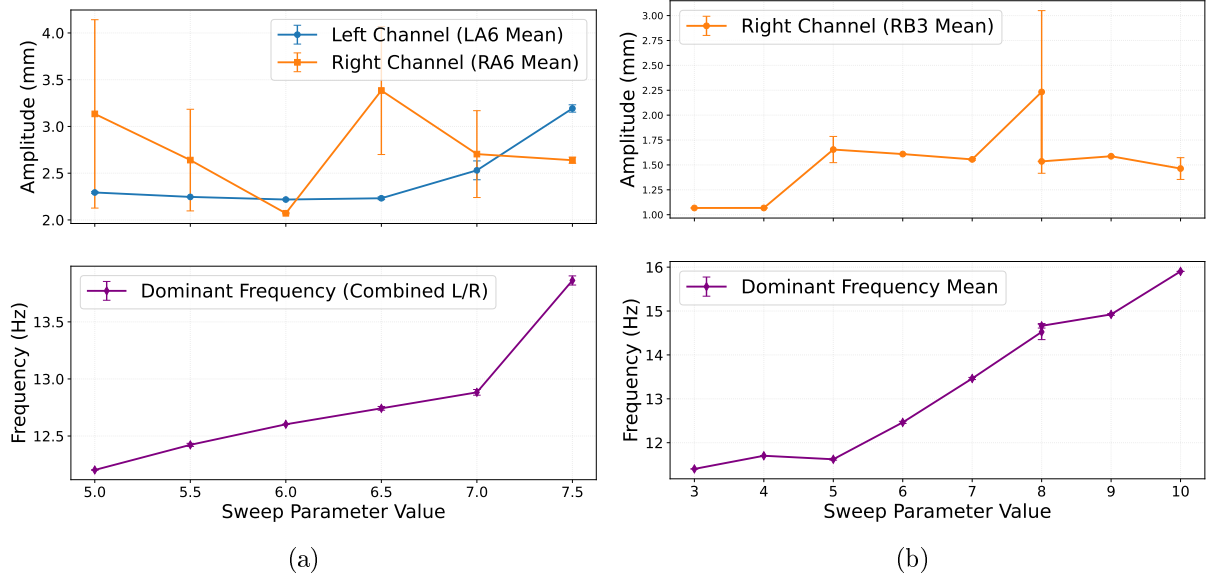


Figure 8: **Experimental data of single unit limit cycles.** Characteristics of the limit cycles, amplitude and frequency, for separate single units. Top: amplitude of limit cycle. Bottom: natural frequency. The x-axis is the activity parameter of odd elasticity of the unit. **8a** First unit. The frequency increases linearly with activity. The amplitude of the left actuator (blue) stays constant at first and then increases. The amplitude of the right actuator (orange) has large error bars, demonstrating its variability. **8b** Second unit. The frequency increases linearly with activity. The amplitude is the same for the first two activity values. For higher activity, the amplitude is larger, but it remains constant throughout.

4 Two non-reciprocal units

In this section we couple two non-reciprocal units with an elastic band.

4.1 Theory

In this section we analyse the classical dynamical equations of 2 non-reciprocal units coupled together via an elastic band. Each unit contains 2 actuators. We use the method of multiple scales to find the complex amplitude equations describing the slow dynamics of the system. Next we switch variables to real amplitude and phase. Finally, we interpret the results in terms of synchronization between the two oscillators.

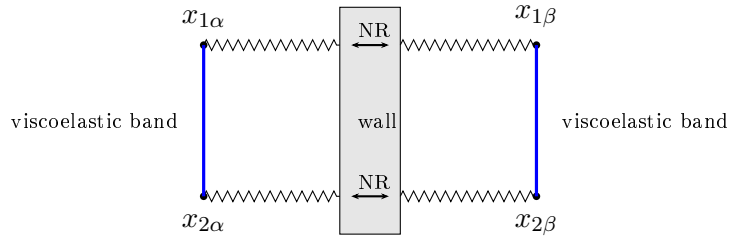


Figure 9: **Schematic representation of two coupled non-reciprocal units.** The actuators are fixed to the wall structure. The actuators on the same side of the wall interact through the viscoelastic band, while the mirror actuators to each side of the wall are coupled non-reciprocally (NR). The displacement of the actuators is perpendicular to the wall.

Since there are 2 actuators per non-reciprocal unit, we have 4 actuators and 8 degrees of freedom in total. We denote the displacements of the actuators in unit 1 as $x_{1\alpha}$ and $x_{1\beta}$, and in unit 2 as $x_{2\alpha}$ and $x_{2\beta}$, as shown in figure 9. The actuators are classical, so we can use Newton's second law to write down the equations of motion:

$$\begin{aligned}
 \ddot{x}_{1\alpha} + \omega x_{1\alpha} &= \epsilon(\omega - \omega_{1\alpha}^2)x_{1\alpha} - \epsilon\gamma_{1\alpha}\dot{x}_{1\alpha} - \epsilon k_{c,1\alpha}x_{1\alpha}^3 + \epsilon k_a^1 x_{1\beta} + \epsilon k_v(\dot{x}_{2\alpha} - \dot{x}_{1\alpha}) + \frac{\epsilon k_b}{2L^2}(x_{2\alpha} - x_{1\alpha})^3, \\
 \ddot{x}_{1\beta} + \omega x_{1\beta} &= \epsilon(\omega - \omega_{1\beta}^2)x_{1\beta} - \epsilon\gamma_{1\beta}\dot{x}_{1\beta} - \epsilon k_{c,1\beta}x_{1\beta}^3 - \epsilon k_a^1 x_{1\alpha} + \epsilon k_v(\dot{x}_{2\beta} - \dot{x}_{1\beta}) + \frac{\epsilon k_b}{2L^2}(x_{2\beta} - x_{1\beta})^3, \\
 \ddot{x}_{2\alpha} + \omega x_{2\alpha} &= \epsilon(\omega - \omega_{2\alpha}^2)x_{2\alpha} - \epsilon\gamma_{2\alpha}\dot{x}_{2\alpha} - \epsilon k_{c,2\alpha}x_{2\alpha}^3 + \epsilon k_a^2 x_{2\beta} + \epsilon k_v(\dot{x}_{1\alpha} - \dot{x}_{2\alpha}) + \frac{\epsilon k_b}{2L^2}(x_{1\alpha} - x_{2\alpha})^3, \\
 \ddot{x}_{2\beta} + \omega x_{2\beta} &= \epsilon(\omega - \omega_{2\beta}^2)x_{2\beta} - \epsilon\gamma_{2\beta}\dot{x}_{2\beta} - \epsilon k_{c,2\beta}x_{2\beta}^3 - \epsilon k_a^2 x_{2\alpha} + \epsilon k_v(\dot{x}_{1\beta} - \dot{x}_{2\beta}) + \frac{\epsilon k_b}{2L^2}(x_{1\beta} - x_{2\beta})^3,
 \end{aligned} \tag{45}$$

where $\omega_{u\sigma}$ is the natural frequency of actuator $\sigma \in \{\alpha, \beta\}$ in unit $u \in \{1, 2\}$, $\gamma_{u\sigma}$ is the damping coefficient, $k_{c,u\sigma}$ is the nonlinear saturation stiffness, k_a^u is the non-reciprocal coupling within unit u , k_v is the viscous coupling of the elastic band, k_b is the coupling strength of the elastic band, and L is the distance between the actuators at rest. The parameter ϵ indicates that all the perturbations, detuning, damping, nonlinear stiffness, non-reciprocal coupling and neighbour coupling, are weak

compared to the harmonic oscillator terms.

After performing the theoretical derivation described in section 2.4, one ends up with differential equations for the amplitude and phase of each of the actuators. For the details of the derivation, see section B. The resulting equation for the radius/amplitude of the actuator $x_{1\alpha}$ is

$$\begin{aligned} \dot{R}_{1\alpha} = & -\frac{1}{2}\gamma_{1\alpha}R_{1\alpha} + \frac{3k_a^1}{2\omega}R_{1\beta}\sin(\phi_{1\beta} - \phi_{1\alpha}) + \frac{k_v}{2}(R_{2\alpha}\sin(\phi_{2\alpha} - \phi_{1\alpha}) - R_{1\alpha}) \\ & + \frac{3k_b}{4\omega L^2}\sin(\phi_{2\alpha} - \phi_{1\alpha})(R_{2\alpha}^3 + R_{1\alpha}^2R_{2\alpha} - 2R_{2\alpha}^2R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha})), \end{aligned} \quad (46)$$

and the equation for the phase is

$$\begin{aligned} \dot{\phi}_{1\alpha} = & \omega_{1\alpha} + \frac{3k_{c,1\alpha}}{2\omega}R_{1\alpha}^2 - \frac{3k_a^1}{2\omega}\frac{R_{1\beta}}{R_{1\alpha}}\cos(\phi_{1\beta} - \phi_{1\alpha}) + \frac{k_v}{2}\left(\frac{R_{2\alpha}}{R_{1\alpha}}\sin(\phi_{2\alpha} - \phi_{1\alpha}) - 1\right) \\ & - \frac{3k_b}{4\omega L^2}\left[\cos(\phi_{2\alpha} - \phi_{1\alpha})\left(\frac{R_{2\alpha}^3}{R_{1\alpha}} + R_{1\alpha}R_{2\alpha} - 2R_{2\alpha}^2\cos(\phi_{2\alpha} - \phi_{1\alpha})\right)\right. \\ & \left. - R_{2\alpha}^2 - R_{1\alpha}^2 + 2R_{2\alpha}R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha})\right]. \end{aligned} \quad (47)$$

These are the equations describing the motion of the first actuator in unit 1. The equations for the other actuators $(x_{1\beta}, x_{2\alpha}, x_{2\beta})$ are obtained in the same way and differ only in some of the indices.

In the amplitude equation (46), there are four terms. The first one is negative, so it reduces the amplitude, yielding a stable origin. It is no surprise that it is the effect originating from friction $\gamma_{1\alpha}$. The sign of the second term depends on the sign of the activity parameter and on the phase difference between the actuators within the same unit $(\phi_{1\beta} - \phi_{1\alpha})$. The activity term can overcome the damping term, and lead to a stable nonzero value of the amplitude. In other words, it generates the limit cycles. It is most influential when the actuators are in quadrature. This agrees with our discussions in section 3. The third term comes from the viscous coupling of the elastic band between units 1 and 2 (actuators 1α and 2α). The units are synchronized when their phases $\phi_{1\alpha} = \phi_{2\alpha}$ are equal. In that case, the term is negative and stabilizes the origin. The fourth term originates from the interactions through the elastic band. If the units are synchronized ($\phi_{2\alpha} = \phi_{1\alpha}$), the whole term vanishes. Then, the equation reduces to that of an uncoupled unit. In other words, if the units synchronize, then they oscillate with their natural amplitude.

The phase equation (47) resembles a Kuramoto-type of model. We can compare the coefficients with experimental data. The first term is the natural frequency of actuator 1α . The second term is a non-isochronous term: the radius of oscillation affects the frequency. The third term is a usual Kuramoto coupling between the two actuators within unit 1 (1α and 1β). As we mentioned before, odd elasticity breaks reciprocity between the actuators. That is why the term is proportional to the cosine instead of the sine of the phase difference. The result is that the actuators within one unit are in quadrature, with phase difference $\phi_{1\beta} - \phi_{1\alpha} = \pm\frac{\pi}{2}$, just like what we saw in section 3.1. The viscous term is another Kuramoto interaction, but now between actuators from different units (1α and 2α) that are interacting through the elastic band. An important feature is that it is proportional

to the sine of the phase difference. Therefore, the viscous coupling seems to be the synchronizing term, as we saw in section 1.4. Also note that the coupling is proportional to the ratio of radii $\frac{R_{2\alpha}}{R_{1\alpha}}$. The last term is a more complicated Kuramoto type of coupling of cosine functions, which is a consequence of the cubic elastic neighbour coupling. Since the cosine is even, the elastic coupling term does not lead to the synchronization of units. This came as a surprise at first. Our primitive model had no viscous interaction terms from the band. Therefore, we arrived at a Kuramoto-like phase equation with no synchronizing term. We realized that such a synchronizing term needed to be imaginary in the Stuart-Landau equations (70). That meant the interaction depended on the velocity rather than on the position of the actuators. Hence, we added the viscous interaction in the equations of motion (45), which indeed was the synchronizing term. The reason is that the elastic term is conservative. By Liouville's theorem, the phase space volume remains constant under time evolution. If the interaction were not conservative, then the phase space volume would shrink towards an attractor. This is precisely the case of the viscous interaction, which dissipates energy. The key idea is that synchronization involves the convergence of many different initial states to the same trajectory. In the case of two coupled units, the phase space has eight dimensions, since there are four actuators and each one has position and velocity. Synchronization converges a region of the phase space to evolve into a single curve. In consequence, synchronization reduces the phase space volume. Therefore, the elastic term cannot synchronize, but the viscous term can.

We can also consider some simplified limits of equations (46) and (47). Actuators 1α and 1β are coupled non-reciprocally, so their phases relock to $|\phi_{1\beta} - \phi_{1\alpha}| = \pi/2$. This means that the non-reciprocal coupling term is maximal in the radius equation, and it is zero in the phase equation. The same thing happens between actuators 2α and 2β . We can also consider the synchronization regime, when the limit cycles of units 1 and 2 oscillate with the same frequency. In the case of complete synchronization (for the α -actuators), both oscillators have the same phase: $\phi_{2\alpha} = \phi_{1\alpha}$. In this limit, the neighbour coupling term in the amplitude equation vanishes, and the neighbour coupling term in the phase equation simplifies:

$$\dot{R}_{1\alpha} = -\frac{1}{2}\gamma_{1\alpha}R_{1\alpha} + \frac{3k_a^1}{2\omega}R_{1\beta} + \frac{k_v}{2}(R_{2\alpha} - R_{1\alpha}), \quad (48)$$

$$\dot{\phi}_{1\alpha} = -(\omega - \omega_{1\alpha}) + \frac{3k_{c,1\alpha}}{2\omega}R_{1\alpha}^2 - \frac{k_v}{2} + \frac{3k_b}{4\omega L^2} \frac{(R_{2\alpha} - R_{1\alpha})^2}{R_{1\alpha}}. \quad (49)$$

Note that the synchronization frequency is not the average of the original frequencies. One would get the average of the original frequencies if the amplitude played no role in the interaction, and only the phase did. One way to show this is to start from a Kuramoto-type of model whose dynamics are:

$$\dot{\Theta}_1 = \omega_1 + K_1 \sin(\Theta_2 - \Theta_1), \quad (50)$$

$$\dot{\Theta}_2 = \omega_2 + K_2 \sin(\Theta_1 - \Theta_2). \quad (51)$$

If we impose the synchronization condition $\dot{\Theta}_1 = \dot{\Theta}_2 := \Omega$, in the case in which the coupling is the

same $K_1 = K_2$, we get that the synchronization frequency is given by

$$\Omega = \frac{\omega_1 + \omega_2}{2}. \quad (52)$$

But in our setup, the amplitude does play a role in the interaction. The larger the amplitude of the first node, the stronger the coupling feels for the second node. In mathematical terms, we expect $K_1 \propto R_2$ and $K_2 \propto R_1$. If we assume that the couplings are different $K_1 \neq K_2$ and massage the expression $K_2\dot{\Theta}_1 + K_1\dot{\Theta}_2 = \Omega(K_1 + K_2)$, we get that the synchronization frequency is given by

$$\Omega = \frac{K_2\omega_1 + K_1\omega_2}{K_1 + K_2}. \quad (53)$$

The crossed indices in the numerator make sense. If K_1 is very large, that means that the influence of oscillator 2 on oscillator 1 is very strong, and therefore the synchronization frequency is closer to ω_2 . On the other hand, if K_2 is very large, that means that the influence of oscillator 1 on oscillator 2 is very strong, and therefore the synchronization frequency is close to ω_1 .

In section B, we obtained what the couplings K_i were. Besides the constant factors, they were proportional to the ratio of the amplitudes: $K_1 = \frac{R_2}{R_1}$. Substituting the couplings in equation (53), we find

$$\Omega = \frac{R_1^2\omega_1 + R_2^2\omega_2}{R_1^2 + R_2^2}. \quad (54)$$

Next we study the linearized dynamics. The spectrum and the time evolution of the eigenvectors are shown in figure 10a. One can see that the spectrum forms a square in the complex energy plane. In other words, pairs of eigenvalues have the same real part and other pairs have the same imaginary part. The eigenmodes whose real energy is the lowest (2 and 3) have in-phase synchronized motion between the 2 oscillators, while the eigenmodes with the highest real energy (1 and 4) have an anti-phase synchronized motion. The imaginary part of the energy also plays a role. Modes with positive imaginary energy are amplified and modes with negative imaginary energy are damped. Therefore, the mode with highest imaginary energy has the most amplification and it will dominate the dynamics. In figure 10a, the highest imaginary energy eigenvalue is the one labeled 2, and its mode corresponds to the actuators 1α and 2α oscillating $\pi/2$ ahead in phase in comparison to actuators 1β and 2β . The mode corresponds to in-phase synchronized counterclockwise oscillations of the limit cycles. This the mode dominates the dynamics. In comparison, mode 3 corresponds to clockwise oscillations. Its smaller value of imaginary energy makes it decay in time and have no appearance on the dynamics. The other mode with positive imaginary energy is eigenmode 0. It also corresponds to counterclockwise oscillations, but the limit cycles are antiphase rather than in-phase. It means that the phase difference between the limit cycles is π . Since this mode has slightly smaller imaginary energy and larger real energy, it is also obscured by the dominating mode. This is what one could expect from intuition, but we still had to check it. By allowing the actuators to oscillate, it is energetically favourable for them to synchronize, and for them to have the same phase.

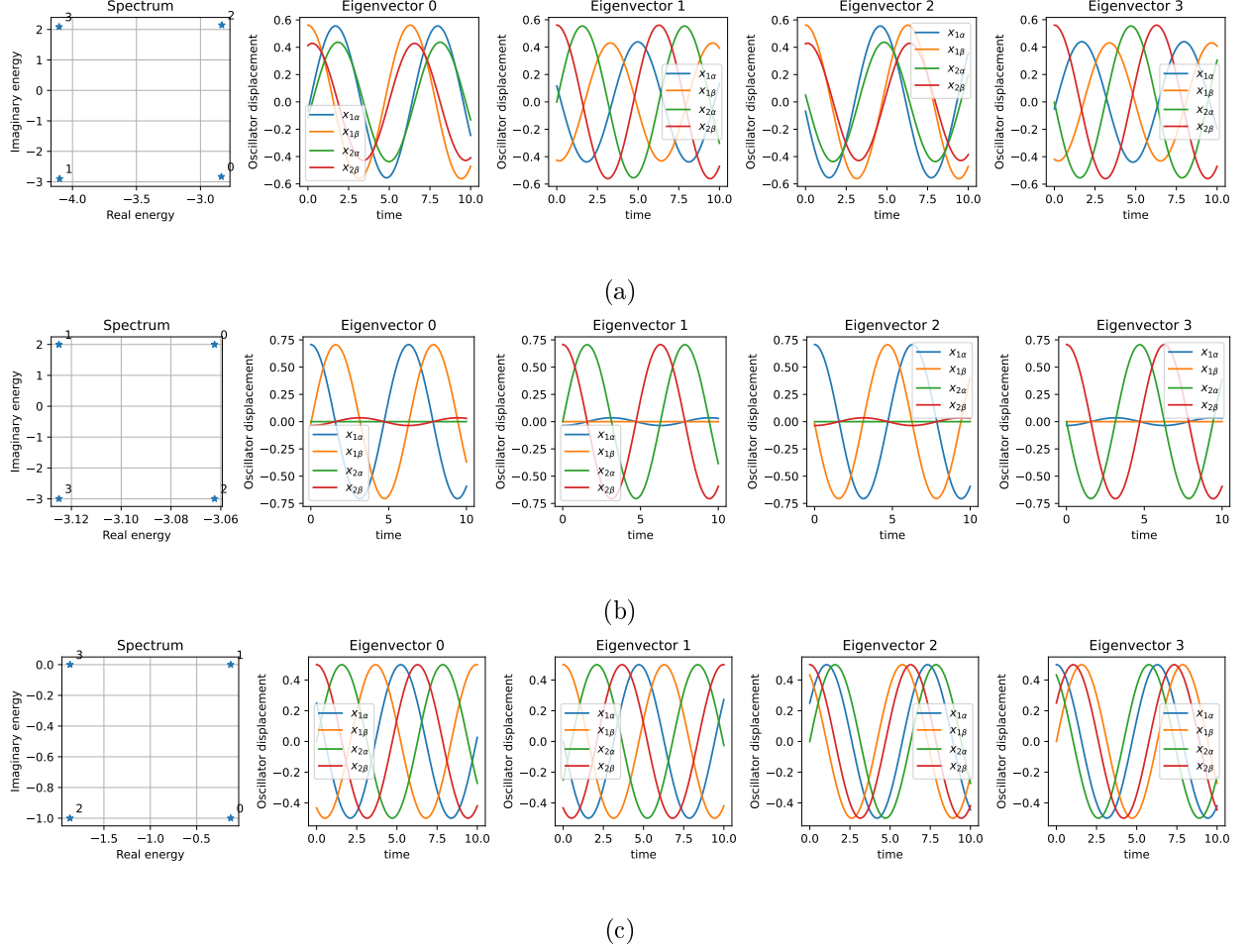


Figure 10: **Spectrum and eigenvectors of linearized system of two coupled non-reciprocal units.** Each row has five panels. The first one is the spectrum in the complex energy plane. The rest are the eigenvectors as a function of time. The displacement of each of the actuator is shown for each eigenvector. The dominant eigenvector in the dynamics is the one with highest imaginary energy. **10a** Two active units with the same chirality. **10b** Two active units with opposite chirality. **10c** One active and one passive units.

We can also compute the spectrum and eigenvectors of two units with opposite chirality. The results are shown in figure 10b. As we said, the dynamics are dominated by the mode with highest imaginary energy. In the case of figure 10b, those are modes 0 and 1. Mode 0 corresponds to large oscillations of the first unit, while the oscillations of the second unit are attenuated. For mode 1 the roles switch, so the second unit has larger amplitude. This means that units with opposite chirality do not oscillate simultaneously with large amplitude. They cancel each other's oscillations. Consider also the case where the first unit is active and the right one is passive. The results for this configuration are shown in 10c. The dominant mode here is number 3. It corresponds to oscillations of the active unit, followed by the passive unit with a phase delay. This is precisely what we expected. The passive unit is dragged along by the active one. The phase delay comes from the time it takes the interaction to travel. That is given by the strength of the viscoelastic couplings k_b and k_v .

4.2 Experiments with same chirality



We can study experimentally what happens when coupling two non-reciprocal oscillators. We place two non-reciprocal units in our wall setup, so 4 linear actuators in total, and the pairs are independent of one another. There is no interaction between the units. We did an experiment in which the non-reciprocal units had the same chirality, but were oscillating with different frequencies above the Hopf bifurcation. One of the nodes was oscillating with a frequency of 13.81 Hz, and the other one with 14.61 Hz. We then added a viscoelastic band coupling the two units together. The result of the coupling was that the nodes synchronized to the same frequency, which was measured to be 14.01 Hz. These measurements showed a phenomenon of the system that I didn't anticipate: synchronization. In fact, the phenomenon is present all throughout the project. After some reading and thinking, the synchronization of non-reciprocal units is understood naturally. Each non-reciprocal unit performs a self-sustained limit cycle. When several of these units are coupled together, they will keep performing the limit cycle but they will also interact with one another. The interactions can cause phase delays and frequency shifts. In most cases, the most favourable mode of oscillation is the one in which they all oscillate with the same frequency and with very similar phase, since this mode minimizes the interactions through the viscoelastic band.

To study synchronization, we considered two units with the same chirality. We kept one of the unit's activity fixed during the whole experiment, $k_a = 7$ for node 4, and we varied the other unit's activity, node 6. That way we only had one parameter to change in the experiment. Figure 11 shows the frequency of the limit cycles performed by the two units for different gain values of node 6. Node 4 data is drawn as circles and node 6 data as squares. For each activity value of node

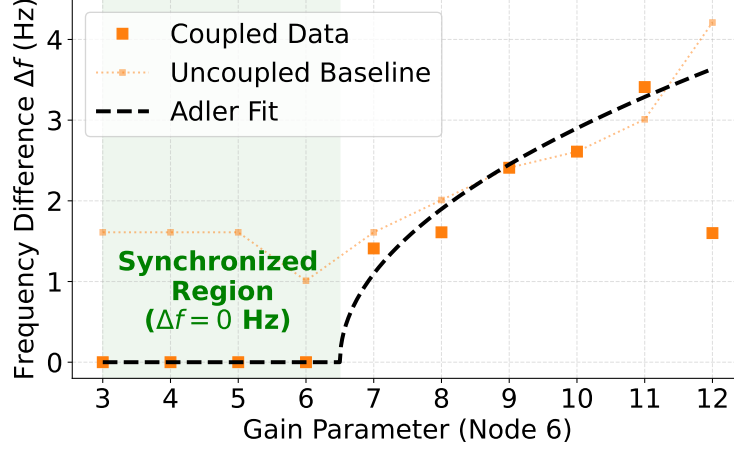


Figure 11: **Synchronization region of two coupled units with the same chirality.** The units are performing limit cycles, and they are coupled together via a viscoelastic band. We fix one of the activity/gain parameters, and we vary the other one in the x-axis. We show the frequency difference between the two units in the y-axis. The region where the frequency difference is zero corresponds to synchronization. The other region where the frequency difference is nonzero corresponds to chaos. The last data point (gain=12) of the coupled data is believed to be a consequence of oscillation quenching.

6, we did 3 types of measurements. The first one was to determine the natural frequency of node 4 when it was oscillating by itself, and node 6 had zero activity. In the plot the data is shown as a soft circles. Since we didn't change its activity throughout the experiment, its frequency is constant. The second type of measurement is the natural frequency of node 6 when node 4 has zero activity. These data points are soft squares. The third type of measurement is both nodes having nonzero activity and coupled together by the viscoelastic band. This is the most relevant type of measurement, and it indicates whether the units synchronize or not. The data points for the frequency in this case are shown as solid or filled circles and squares. Since each unit is made out of two actuators, denoted L and R, there are double the number of data points. However, L and R are coupled non-reciprocally to form the limit cycles, and that forces them to already oscillate with the same frequency. Then, the corresponding L and R data points coincide.

Looking at the whole figure 11, there are two regions. For high activity parameter of node 6, the units oscillate with different frequencies. Therefore, the units are not synchronized in that region. For low activity values of node 6, in the range $k_a \in [3, 6]$, the units do synchronize. The results are consistent with the theory. We saw that the synchronization of two oscillators can be described by the Adler equation. If the coupling was larger than the detuning, then the oscillators synchronized. If the coupling was smaller than the detuning, then they didn't synchronize. These two regions are clearly visible in figure 11. For $k_a \in [3, 6]$, the oscillators synchronize, which means that the coupling is larger than the detuning. For $k_a \in [7, 12]$, the oscillators don't synchronize, which means that the detuning is larger than the coupling. However, the Adler equation cannot explain the whole plot. We observe that the detuning is the same for $k_a = 3, 4, 5, 7$, but the units synchronize for

$k_a = 3, 4, 5$ and they don't synchronize for $k_a = 7$. What has changed between the measurements? It is important to keep in mind that the only parameter we control is the activity of one of the units. As we saw in section 3.3, the activity of a unit affects both its natural frequency and its amplitude of oscillation. The detuning is not a variable we control directly, but we can control it indirectly through the activity. In experiments, the detuning can be easily measured by comparing the natural frequencies of the two units when they are uncoupled. One might think that the coupling is constant, because the viscoelastic band remains unchanged in the experiment. However, we saw in section 3.3 that the amplitude of oscillation of a unit depends on its activity. And in section 3.1, we saw that the Kuramoto-like couplings between two non-reciprocal units depends on the amplitudes. Therefore, the coupling in the system is not constant and depends nontrivially on the activity. The Adler equation is derived by taking the phase dynamics approximation, which assumes that the amplitudes are constant. In the experimental setup of non-reciprocal actuators, the approximation does not hold if we change the activity, and so the results are not expected to be fully explained by the Adler equation.

We observe that the frequency of node 6 grows linearly with its own activity in the uncoupled case. In the coupled case, after the units stop synchronizing, the coupled frequency of node 6 follows closely the uncoupled one. Therefore, after transitioning to the non-synchronized region, each unit oscillates with a frequency similar to the uncoupled one. The small coupling, in comparison with the detuning, has a small effect on the oscillation frequency in this region. We can also observe that as the detuning becomes larger, the effect of the coupling becomes smaller, and so the units oscillate with their natural frequencies. This observation can also be explained by the Adler equation, whose relevant parameter was the ratio of coupling to detuning. As the detuning increases, the effective coupling decreases. Nevertheless, for $k_a = 12$ something different happens. The frequency of node 6 drops substantially when coupling it to the other node. The data point might be a symptom of a phenomenon called oscillation quenching. It could also be caused by the actuators oscillating erratically due to the large value of activity. In fact, we didn't take measurements for higher activity because we were afraid the actuators were going to break.

We wanted more reliable data, so we carried out another experiment in which we repeated each measurement 3 times. The actuators used were different to the previous experiment, so one cannot compare the specific details of the results. But using different actuators allows to distinguish what results are general and robust, and which are not. The results are shown in figure 12. Again, one of the units had a fixed activity, in this case it was node D1 with $k_a^{D1} = 6$ (shown in blue). The other unit's activity was varied, in this case it was node F1, shown in red. The soft dashed lines label the uncoupled data, while the solid colourful lines label the coupled data. Focusing first on the uncoupled frequencies, we observe that the frequency of unit F1 grows linearly when changing its activity. This is consistent with the results from the previous section and a good sanity check. Nevertheless, there is another sanity check that is not fulfilled. Since we are only changing the activity of node F1, we expect the uncoupled frequency of node D1 to remain constant. Figure 12 shows that it is not constant. Even though the measurements have large error bars, the frequency

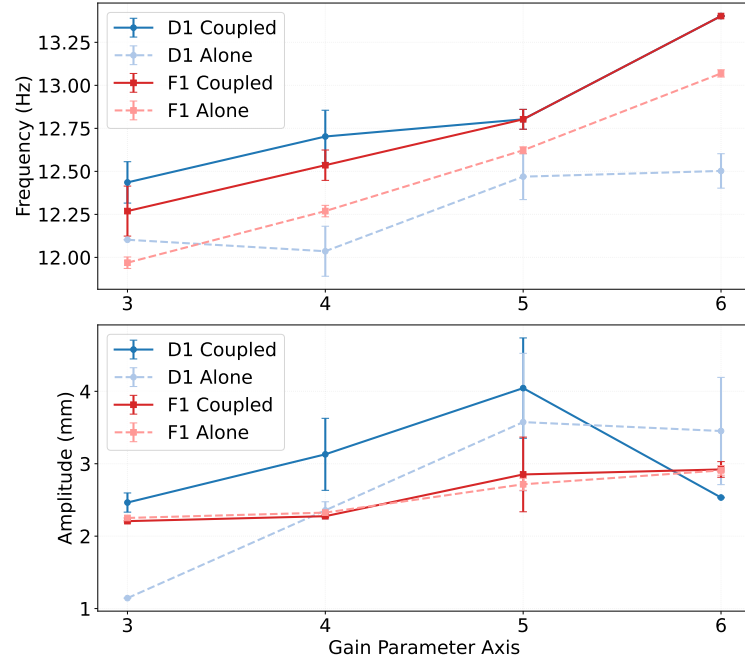
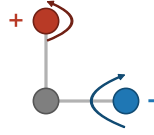


Figure 12: **Experimental data of two coupled units with the same chirality.** The uncoupled/natural data is displayed by dash, dimmed lines. The data taken when they are coupled is displayed in solid, colourful lines. Different units are labeled with different colours. The x-axis is the activity/gain parameter of the red unit (F1). The activity of the blue unit (D1) is fixed to a constant value of $k_a^{D1} = 6$. **Top:** frequency of oscillations. **Bottom:** amplitude of oscillations.

fluctuates and some results are incompatible. These fluctuations illustrate the poor quality of the actuators used in the experiment and the poor reproducibility of results. Although the results are not precise, making a quantitative analysis useless, we can still make some qualitative observations and conclusions.

The first observation is that the units synchronize in most of the measurements. For $k_a^{F1} \in \{5, 6\}$, the data points of both units coincide, so they clearly synchronize. For $k_a^{F1} \in [3, 4]$, the data points do not coincide. Note that the data points do not coincide because in one of the three trials, the units were not synchronized and oscillated with different frequency. Thus, it could be that the units did synchronize in other trials, since the error bars are compatible. And that is precisely what one finds if one looks at individual trials. For $k_a^{F1} = 4$, the units synchronize in two trials and they don't synchronize in one of them. For $k_a^{F1} = 3$, the units synchronize in one trial and they don't synchronize in two of them. Even in trials where the units did not synchronize, one can find periods of synchronization. And also, in trials where the units did synchronize, one can find phase slips. These observations can be seen in figure 27 in the appendix C, which shows the phase difference between the two units.

4.3 Altermagnet unit cell



We modified the previous configuration to units with opposite chirality. We also extended the system by adding a passive site and a mixture of horizontal and vertical couplings. The motivation for this geometry is that it is precisely the unit cell of the 2D Lieb lattice. We can consider coupling two units in an L shape, with opposite chirality, and adding a passive site in the vertex of the L. The passive site is mediating the interactions between the non-reciprocal units. In the experimental results of figure 13a, we fixed the gain of the negative-chirality unit (B5) to be $k_a^- = -7$, and we changed the gain of the positive-chirality unit (C4). The results for the frequency and amplitude are shown. Each data point is the average of 3 measurements. The error bars are the standard deviation of the 3 measurements. As before, the soft lines indicate the motion when the units are uncoupled, and the solid colourful lines when they are coupled. What one observes in the frequency plot is that the units don't synchronize for any of the activity values. Even at the gain value of $k_a^+ = 7$, when their uncoupled frequencies are similar, further analysis shows that the units don't synchronize. For $k_a^+ = 8$, the viscoelastic band coupling brings the frequencies closer together in comparison to the uncoupled case. This suggests that when the uncoupled frequency difference is small, there is an interaction between the units. The interaction tends to synchronize the units, but it doesn't succeed in doing so. We infer that synchronization doesn't happen when the units have opposite chirality. We run some simulations to test this hypothesis.

The observation can be explained with the Adler equation (6). The first order approximation of the viscoelastic band coupling term is $\sim K \sin \Delta\phi$, as is shown in equation (47), which is derived in appendix B. As the term originates from the viscoelastic band, it is expected to be equal for same and opposite chirality configurations. In contrast, the detuning significantly changes from one configuration to the other. In the case of two units with the same chirality, they have similar natural frequency, though not exactly equal. Therefore, the detuning is notably small: $\Delta\omega = \omega_2 - \omega_1 \ll \mathcal{O}(\omega_i)$. But in the case of two units with opposite chirality, one unit oscillates clockwise and the other counterclockwise. We can assign a sign to each of the directions, according to the chirality. This means that the natural frequencies actually have opposite signs, but are of the same order of magnitude. This means that the detuning is large: $\Delta\omega = \omega_2 - \omega_1 \sim 2\omega_i$. The Adler equation predicts that when the detuning is larger than the interaction coupling, then the units do not synchronize. This is precisely what we obtain in the experiments. If the units have the same chirality, the detuning is small, and they synchronize. If the units have opposite chirality, the detuning is large, and they don't synchronize.

What we observed in simulations of two units with opposite chirality is that the units only synchronize if all the parameters are exactly the same. If the condition is not satisfied, then the units don't synchronize. The condition remains true for small and large values of gain. If the gain difference between the two units is not small, then the simulations show that the frequencies of oscillation in the coupled case are almost identical to the uncoupled case. The coupling has a very small effect and each unit follows their natural motion. If the gain difference is small, then the frequencies of oscillation drop considerably with respect to the uncoupled ones. In that case, both units oscillate with similar frequency, but not exactly the same. Also, the amplitude of oscillations in this case is reduced substantially with respect to the uncoupled case.

We mentioned before that the experimental units are notably inhomogeneous in their properties. We model this observation in the simulation by introducing quenched disorder in all of the parameters. And we saw in the simulations the units with opposite chirality only synchronize if all the parameters are exactly the same. However, in the experiment we can control only the gain parameter, the rest of the parameters are fixed and unique for each unit. By the rest of parameters, we mean the linear, cubic spring constants and the damping coefficient. Therefore, even if we set exactly the same gain for the two units, the other parameters will be different, and that prevents the synchronization. In conclusion, units with opposite chirality don't synchronize in general.

The conclusion is also supported by the theoretical predictions. We can describe the phase difference of the two units as an Adler equation with higher order terms, which we can neglect in a first approximation. The condition of two units with opposite chiralities means that one unit has a positive oscillation frequency, i.e. the phase grows monotonically, and the other unit has a negative oscillation frequency, in the sense that the phase decreases monotonically. Therefore, the frequencies of the two units are roughly opposite of each other: $\omega_1 \sim -\omega_2$. Then, the detuning for opposite chirality is large $\Delta\omega \sim 2\omega_1$. It is much larger than any detuning encountered in the configuration of

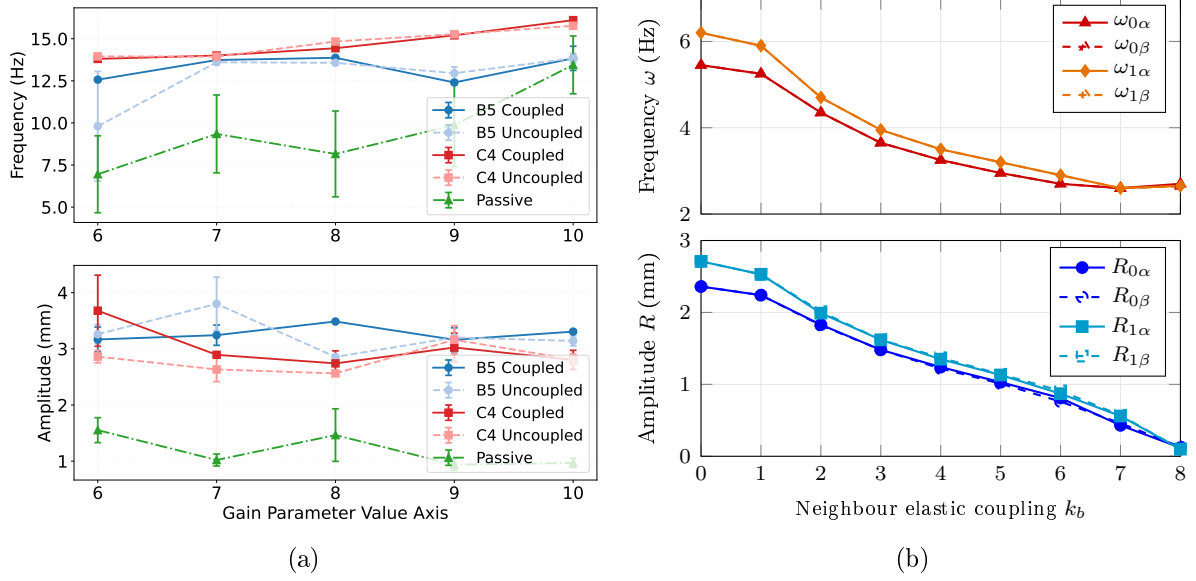


Figure 13: **Experimental and simulation data of two coupled units with opposite chirality.** The configuration includes a passive site and both vertical and horizontal couplings, to reproduce the altermagnet unit cell. **13a** Experimental data of frequency (top) and amplitude of oscillations (bottom). The uncoupled/natural data is displayed by dashed, dimmed lines. The data taken when they are coupled is displayed in solid, colourful lines. Different units are labeled with different colours. The x-axis is the activity/gain parameter of the red unit (C4). The activity of the blue unit (B5) is fixed to a constant value of $k_a^{B5} = -7$. **13b** Simulation data of frequency (top) and amplitude (bottom) of oscillations, of individual actuators in each unit, as the elastic coupling strength is varied. The subscripts 0 and 1 denote the different units. The superscripts α and β denote the two actuators within each unit.

units with the same chirality. Since the coupling strength is similar to the one found in configuration of units with same chirality, the system with opposite chirality is in the region where $\Delta\omega > K$. The Adler equation predicts no synchronization, which agrees with the experiments and simulations.

One might ask what the effect of adding the passive site and having both vertical and horizontal couplings were. These two modifications do not change the phenomena observed, hence the alter-magnet unit cell has the same behaviour as two coupled units with opposite chirality. Analogously, the behaviour of a ferromagnet unit cell is qualitatively the same as the one of two coupled units with the same chirality. Thus, we skip the analysis of those configurations. The modifications do change the details of the results. For example, adding the passive site introduces more degrees of freedom and modes of oscillation that absorb energy from the active units. It makes the interaction strength between active units effectively smaller. As a consequence, if one couples two units with same chirality and introduces a passive site in between, the system requires a higher activity for the nodes to synchronize. At low values of activity, the passive site damps the displacement of the active ones, and the interaction is too small for there to be synchronization. Having both vertical and horizontal couplings of the viscoelastic band had no effect on the qualitative features of the system. The viscoelastic band still mediated the interactions, and change of direction did not matter. However, we did observe slight differences in the results when coupling units in different directions. In order to study this asymmetry and the combination of different interactions, we consider in the next section a cross cell.

4.4 Simulations

We consider two units with opposite chirality. We fix the activity of the units to $[7, -8]$, and saw the effect of changing the neighbour coupling k_b in the dynamics. There are three regions. If $k_b \sim \mathcal{O}(0.1)$ is small, then the interaction is very small and the oscillators move independently. If $k_b \sim \mathcal{O}(1)$, the amplitudes of oscillations are modulated periodically. As the coupling is increased further, the amplitude decreases to zero.

Next, we fixed the neighbour coupling to be $k_b = 10$ and changed the values of activity of both units. If the activities were exactly opposite one of the other, then the oscillations cancel each other out. The displacements are very small. To compensate for it, we increase the activity of both units in magnitude, but keeping them opposite to each other. What we find are still small oscillations, but of same order of magnitude to the previous simulations. Surprisingly, the units synchronize. However, the synchronization is not robust under small disorder in the other parameters. This explains why in the experimental setup, in which there are imperfections in the actuators, we did not observe synchronization between two units with opposite chirality. If the activities are slightly different in magnitude, the simulation shows that the units do not synchronize. When the units are uncoupled, the limit cycles have different natural frequencies and different natural amplitudes. When coupling them together, the dynamics alternates through periods of large and small amplitudes. This is

shown as a modulation of the oscillations of individual springs. The modulation is periodic, but in a timescale much larger than the period of oscillation. The modulation is caused by the phase difference between the two units. If the phase difference is far from being zero, then the oscillations are not canceled much, so the units oscillate with considerably large amplitude. But when the phase difference between the units is close to zero, then the cancellation is much greater, and the amplitude is reduced. Since they do not synchronize, the phase difference is always increasing in time, and therefore we get alternating periods of large and small amplitude. The period of the modulation is then related to how different the magnitudes of activity are. If the activities are just slightly different, the period of modulation is very large, around $T_{\text{mod}} \sim 30\text{ s}$, while the fast oscillations are order $T_{\text{fast}} \sim 0.1\text{ s}$. For arbitrary different activities, we observe a modulation of $T_{\text{mod}} \sim 10\text{ s}$. If one keeps making the activity difference even bigger, one actually encounters another case, besides synchronization, when two units with opposite chirality have no modulation. It happens when the activities are so different between each other that the natural frequencies are multiples of each other.

5 1D chain

5.1 Theory

Before going to the Lieb lattice in 2D, we wanted to get intuition on a 1D system, a chain, both classically and quantum mechanically. Classically just means considering the sites as oscillators coupled with springs, and quantum mechanically as sites with hopping amplitudes.

Let's first consider the quantum case. We studied the Hatano-Nelson chain [16], with some added twists. First, we stacked 2 chains on top of each other. We did this to have a two-level system per site and to be able to try different symmetries. And second, we added non-hermiticity, in two ways. The first way is adding an imaginary onsite potential. The second way is adding non-reciprocal hoppings. The spectrum and the eigenstates are shown in the annex. Note that as we are now dealing with a non-Hermitian system whose behaviour is going to be sensitive to the boundary conditions we choose. As a consequence, we have to study the spectrum and eigenstates with both periodic and open boundary conditions.

One of the things that stands out from the results is the symmetry that the spectrum manifests. We can observe that the spectrum displayed in the complex energy plane is symmetric with respect to the real and imaginary axes.

We now show the analytic results of the theoretical approach in the specific case of a one-dimensional chain. For simplicity, we assumed that the neighbour coupling between units is linear. We already know that the two actuators making up the unit will have a phase difference of $\Delta\phi = \pm\frac{\pi}{2}$, and so the odd elastic term in equation (47) will vanish. By taking the mentioned simplifications

in equation (47), we arrive at the equation describing the phase dynamics of one quadrature-phase unit with neighbours:

$$\dot{\phi}_i = \omega_i + \frac{3k_c}{2\omega} R_i^2 + \sum_{\langle j \rangle} \frac{k_v}{2} \left(\frac{R_j}{R_i} \sin(\phi_j - \phi_i) - 1 \right) + \sum_{\langle j \rangle} \frac{k_b}{2\omega} \left(1 - \frac{R_j}{R_i} \cos(\phi_j - \phi_i) \right). \quad (55)$$

For very long chains of identical oscillators, we can take the continuum approximation. Since the units have the same activity, we can assume that their amplitudes are fixed to the same value. The only degree of freedom left then is the phase. Since we are considering a one-dimensional chain, each unit has two nearest neighbours: one to the left and one to the right. Thus, the sums in equation (55) are made up of two addends. We expand in Taylor series only the last two terms, since they are the ones that include phases from different units. Using $\sin \Delta\phi \simeq \Delta\phi$, we simplify the viscous neighbour coupling term to be $\frac{k_v}{2} (\phi_{i-1} + \phi_{i+1} - 2\phi_i - 2)$. This is precisely the discrete second derivative of ϕ , when one takes the distance between units l to zero. And using $\cos \Delta\phi \simeq 1 - \Delta\phi^2$, we simplify the elastic neighbour coupling term to be $\frac{k_b}{2\omega} [(\phi_{i-1} - \phi_i)^2 + (\phi_{i+1} - \phi_i)^2]$. This is the gradient of the phase squared. The resulting continuum equation is the PDE:

$$\frac{\partial \phi}{\partial t} = \underbrace{\omega + \frac{3k_c}{2\omega} R^2 + k_v}_{\Omega} + \frac{k_v l^2}{2} \nabla^2 \phi + \frac{k_b l^2}{2\omega} (\nabla \phi)^2 \quad (56)$$

The first three terms are constant, the fourth is diffusive and the fifth one is nonlinear. Formula (56) is the deterministic Kardar-Parisi-Zhang equation [17]. If equation (56) only had the first constant terms, all the phases increase homogeneously in time: $\phi(x, t) \approx \Omega t$. This is expected, as the units are autonomous oscillators. We now focus on the effect of the diffusive term. We disregard the rest. By expanding in Fourier modes, we find that the phase field can be written as

$$\phi(x, t) \approx \sum_k c_k e^{-\frac{k_v l^2}{2} k^2 t + i k x}. \quad (57)$$

We notice that each mode decays with a time constant (or mean lifetime) $\tau = \frac{2}{k_v l^2 k^2}$. The decay rate depends on the wavevector. For short wavelength modes, the decay is very fast. Longer wavelength modes have a slower decay rate. What we expect to happen in the dynamics is that the phase field becomes progressively smoother as time evolves. The short wavelength fluctuations die before the long wavelength fluctuations. At the end, only the $k = 0$ mode survives, in which the phase field is constant in space. The diffusive term synchronizes the oscillators locally. The observable result is that the wavelength of the system increases in time. The end result is the synchronization of the whole chain. This phenomenon is called wave coarsening [8]. The nonlinear term has the opposite effect. It makes the gradients grow, so it can lead to the formation of kinks or fronts. The deterministic KPZ equation can be transformed into the heat equation by a Cole-Hopf transformation. Therefore, we expect diffusion to dominate for long times. In fact, the steady state solution of the heat equation with open boundary conditions is homogeneous. Since our field is the phase, this is equivalent to synchronization.

There is a synchronization timescale, or in other words, a diffusion timescale for the phase. It is given by the slowest decaying mode: $k_1 = \frac{2\pi}{L}$, where L is the length of the whole chain. We obtain a synchronization timescale of

$$\tau_{\text{sync}} = \frac{L^2}{2\pi^2 k_v l^2}. \quad (58)$$

We observe that it scales with the square of the system size. In consequence, longer chains take longer to synchronize.

5.2 Experiment

We prepared an experimental configuration of a one-dimensional chain. Since each unit has two actuators, there are actually two chains, one to each side of the wall. The configuration had 7 units in total, 3 of which were active.

At first, we set the active units to have the same chirality and studied the regime above the Hopf bifurcation. The spatio-temporal plot is shown in figure 15a. Since the different colours indicate opposite displacements, the figure shows that there are oscillations in both the left and right actuators. If one plots the state space of each unit, the trajectory are the limit cycles we have discussed. The important feature of this figure is that the red and blue stripes are roughly vertical. That shows that the limit cycles in the steady state are phase-locked. They are oscillating with the same frequency and practically in-phase. There are slight differences in the phase, and the reason is that the active units drag the passive ones along with them. Therefore, the passive units (1,3,5,7) have their oscillations slightly delayed with respect to the active ones. In summary, a one-dimensional chain with the same chirality synchronizes, even if passive sites are added in between.

Again, the reason why we observe synchronization is because we are in the regime after the Hopf bifurcation. The units trace out limit cycles, and that means that the oscillators are autonomous. Before the Hopf bifurcation, the units exhibit damped oscillations. The one-dimensional chain can host waves that propagate and attenuate. This regime was difficult to access experimentally, so we studied it using simulations (see section 5.3).

We also carried out experiments where one of the units, specifically the second one, had the opposite chirality. This change results in a considerable reduction of the amplitude of oscillations of the actuators. In order to counteract the effect, we increased the activity of the opposite-chirality unit. The corresponding spatio-temporal plot is shown in 15b. In contrast to the chain with the same chirality, the system does not have an homogeneous appearance. In the figure, the actuators of the first three units display strong colours, while the rest of the figure gradually becomes white. Stronger colours mean that the amplitude of oscillations is large and white means that the amplitude is close to zero. As the second unit has the highest activity, we can think of it as the leader of the

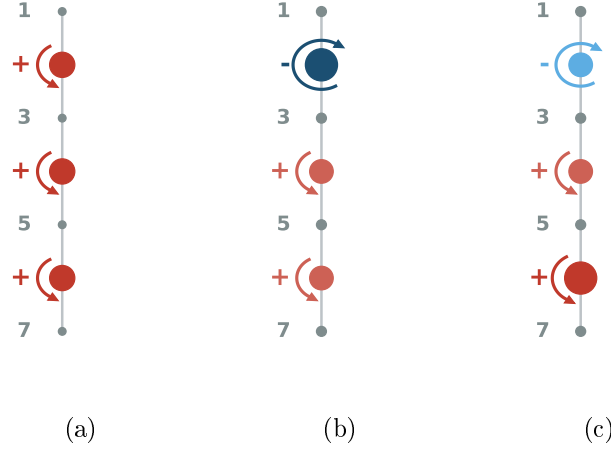


Figure 14: **Diagrams of the three configurations considered for the experiments in a 1D chain of units.** The red circles denote active units with positive chirality. The blue circles denote units with negative chirality. The grey dots denote passive units. The strength of the activity parameter is highlighted by bigger and more intense circles. **14a** Chain with positive-chirality units, and passive sites in between. **14b** The second unit has negative (opposite) chirality with high activity. **14c** Similar configuration to 14b, but now the sixth unit has the highest activity.

oscillations. The passive neighbour unit 1 follows the motion of the leader. However, the units with positive chirality cannot synchronize to the motion of the negative chirality unit. As they have lower activity in comparison, their oscillations are attenuated. In fact, the right actuators of units 3 and 4 are oscillating anti-phase with respect to unit 2.

We can carry out another experiment in which unit 6 has the highest activity. The spatio-temporal plot is shown in 15c. In this case, the amplitude is large for unit 6, small for units 5 and 4, and practically zero for the rest. Since the oscillations are damped, we observe diagonal strips of red and blue, rather than the vertical strips of panel 15a. The diagonal stripes represent that the phase difference between the units is nonzero. Unit 6 generates waves that propagate through the chain and attenuate at the same time.

All things considered, we observed spin-momentum locking in the one-dimensional chain. Placing units with the same chirality leads to synchronization with a large oscillation amplitude. In contrast, placing units with opposite chirality reduces significantly the amplitude, prohibiting the propagation of waves.

We started by building 2 chains of actuators, one chain to each side of the wall. Actuators that are symmetric with respect to reflection of the wall can be coupled non-reciprocally. In general, we won't choose all actuators to be non-reciprocally coupled for geometric reasons. The actuators within the same chain are coupled mechanically via a viscoelastic band.

In the first set of experiments, we had 3 PCBs that would control 3 non-reciprocal units. We had

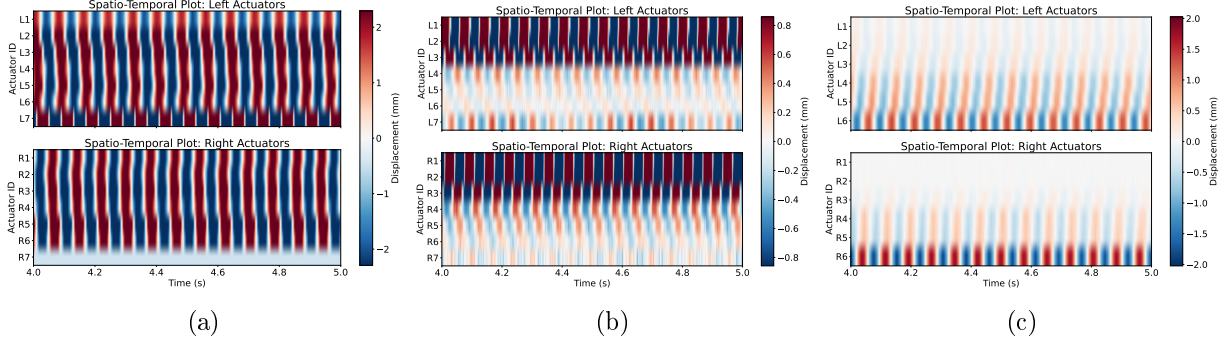


Figure 15: **Spatio-temporal plots of the 1D chain.** Each panel corresponds to the diagrams in figure 14. The top plot shows the displacement of the left actuators (one side of the chain), while the bottom plot shows the displacement of the right actuators (other side of chain). **15a** Kymograph of the 1D strip with all nodes having the same chirality. **15b** Kymograph of the 1D strip with node 2 having highest activity and opposite chirality to nodes 4 and 6. **15c** Kymograph of the 1D strip with node 6 having highest activity.

the option to add passive springs in between the active ones. We wanted to observe the same phenomena we had seen in the simulations, namely, spin-momentum locking. At first, all active nodes were set with the same chirality. The spatio-temporal plot (or kymograph) of figure 15a shows that the nodes synchronize. However, when we flip the chirality of one of the nodes, the synchronization is lost. In fact, the actuators' oscillation attenuate as can be seen in figure 15b. This is a demonstration of the spin-momentum locking. If the nodes have the same chirality/pseudospin, then they synchronize and can be used to send waves (maybe). If they have opposite chirality/pseudospin, the oscillations die and there is no wave propagation.

5.3 Simulations

We also run simulations of the 1D chain of actuators. One important difference of the simulations with respect to the experiment, is that we assume the neighbour coupling is linear instead of cubic. The geometric structure of the chain is the same as the experimental one. There are two chains of linear actuators, one on top of the other, that oscillate perpendicular to the direction of the chain. Each actuator on one chain has a corresponding partner in the other chain, and they are both on the same chain site. The actuator partners interact non-reciprocally, while the closest neighbours within the chain interact mechanically through the viscoelastic band. The parameters of the simulation are:

- Number of nodes
- Onsite stiffness / natural frequency
- Onsite cubic nonlinearity

- Damping coefficient
- Activity: odd elasticity coupling
- Neighbour linear elastic coupling strength (simulates the elastic band)
- Mass of actuators
- Distance between neighbours

We set the initial condition to be a kick in one side of the chain. If the activity is low, then the kick doesn't propagate and it decays. There is no formation of limit cycles because the damping dominates. Therefore we need sufficiently high activity to reach limit cycles. There also needs to be sufficient neighbour coupling so that the perturbation propagates through the chain. Otherwise only the edge node oscillates and the rest remain static.

Simulations allow the study of longer one-dimensional chains that were not accessible experimentally with our setup. We considered units with the same chirality, no passive neighbours in between, with small disorder in individual parameters, and random initial conditions for each actuator. That meant the initial state of the chain was erratic. As the oscillators have same chirality and similar activity, they had a tendency to synchronize and to oscillate in phase locally. Therefore, what we see as time evolves in figure 16 is the phenomenon of wave coarsening. The small-wavelength modes generated by the initial conditions disappear faster than the long-wavelength modes. As time evolves, the chain becomes smoother while still possessing global undulation. The final state of the chain is flat, which is the synchronized steady state. All the units oscillate in unison. The figure shows the transient state from the initial state of the units with different initial conditions to the steady state for which all the units are synchronized. This process happens in all of the experiments we performed. However, it happens in a very short time, within a second. The phenomenon described is robust under changes in initial conditions and under disorder in parameters like activity, linear stiffness and damping.

We considered several chains with different number of units. For chains with 15 units, we observed synchronization after roughly 10×10^3 s, like figure 16a shows. But when increasing the system size to 100 units, we observed no synchronization. The reason is that the synchronization timescale (58) scales with the square of the system size. Thus, it takes a factor $\left(\frac{100}{15}\right)^2 \approx 44$ times longer for the 100-unit chain to synchronize compared to the 15-unit chain. We verified roughly the scaling by considering chains with 10 and 20 units. The synchronization timescale was roughly four times larger.

We included disorder by considering gaussian distributed contributions to a base value. The amount of disorder was set by choosing a specific value of the standard deviation of the Gaussian distribution, for different parameters. For low disorder, we observe synchronization. If one increases the disorder in the activity parameter, while keeping the neighbour coupling relatively low, then

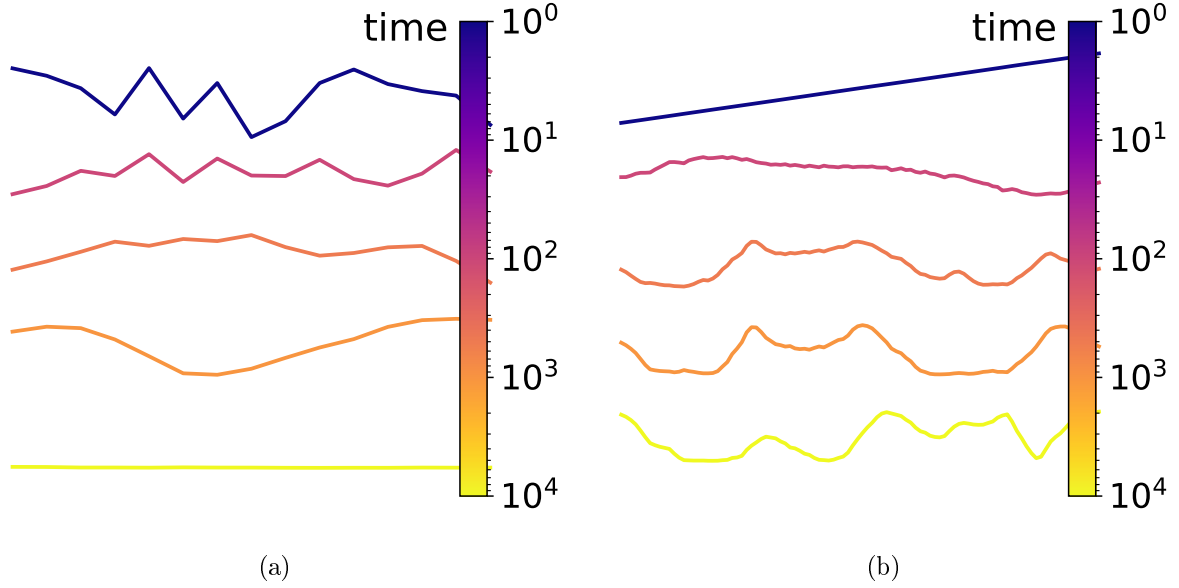


Figure 16: **Wave coarsening in a chain with of non-reciprocal units.** The displacement of one side of the chain is shown in the vertical direction as time evolves. **16a** Chain of 15 units with random initial conditions. **16a** Chain of 100 units with a constant slope as initial condition, which corresponds to a kink in the phase.

one observes several separated regions of synchronization. The units with lowest activity oscillate with small amplitude and their phase is appreciably delayed with respect to their neighbours with higher activity, although they remain synchronized. As the activity disorder keeps increasing, the regions of synchronization become smaller, and some regions are made up of a single unit.

6 Cross cell

So far we have studied and understood the dynamics of a single active unit and the interactions between two active units in different configurations. Before analysing the whole 2D Lieb lattice, we thought it would be wise to study the dynamics of a cross or diamond cell. The cross cell is made out of four active units and one passive unit, shown in figures 22 and 18. The passive unit lies in the center of the cross, and it is connected up, down, left and right to the four active units. We expect the active units to carry out limit cycles. The interaction between the active units is mediated by the passive unit. In some sense, it is like the altermagnet or ferromagnet unit cell, but with four active units instead of two. The main difference with the unit cell configurations is the passive site is mediating several interactions at the same time. We wondered if the passive unit could synchronize all the active units, if it could synchronize only a subset of the active units, or if it wasn't able to synchronize them. As we did before, we can consider two different configurations depending on the sign of the chiralities: ferromagnet and altermagnet.

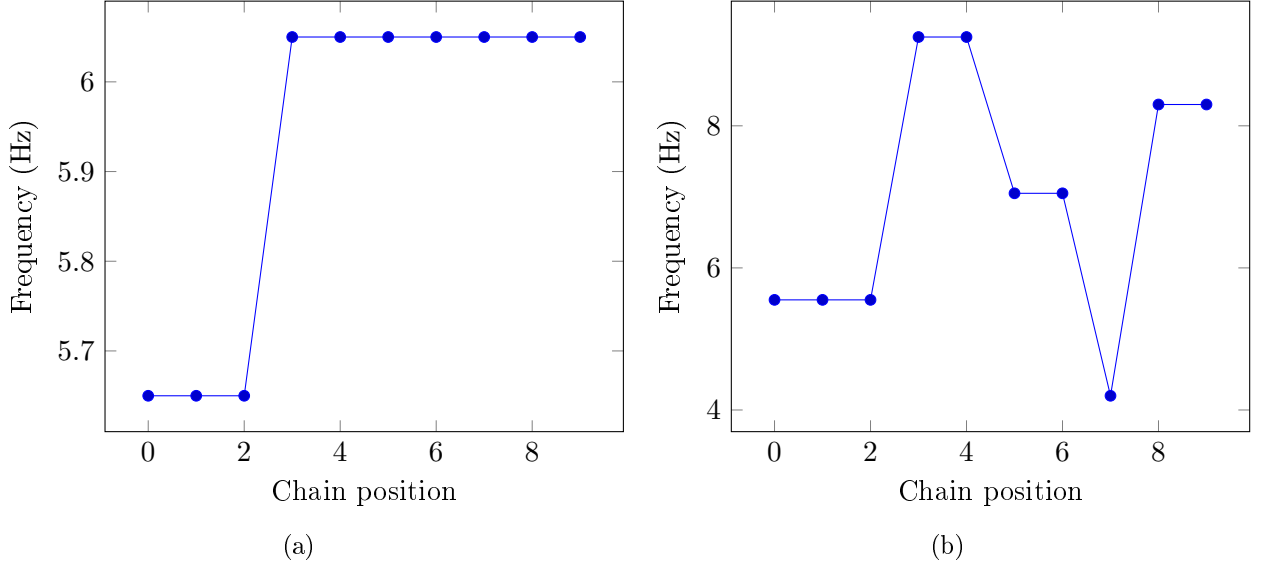


Figure 17: **Separate synchronization regions in a chain with quenched disorder.** The frequency of each unit is shown for two values of disorder in the activity parameter. The disorder is added as gaussian distributed noise with standard deviation $\sigma(k_a)$. **17a** The disorder in activity is $\sigma(k_a) = 2$. The chain has two synchronization regions. **17b** The disorder in activity is $\sigma(k_a) = 4$, higher than the previous one. The chain has five synchronization regions. Increasing the disorder yields the breaking of synchronization into smaller regions. The other parameters of the simulation are: $\mu(k_a) = 7$ $k_b = 3$.

6.1 Ferromagnet cross cell

The ferromagnet cross cell (figure 18) is made out of four active units with the same chirality, and one passive unit in the center. Using the observations made in the previous sections, we expect the four active units to synchronize to the same frequency, and oscillate in phase. In order to observe it, we tuned the activity parameters of the units so that they would have similar natural frequencies. However, the experimental results shown in figure 19 do not support synchronization. We carried out three experimental runs with the same exact activity parameters, and found different results. Figure 19 shows the phase difference with respect to the Left unit, as time evolves. In the first experimental run, the units Left, Center and Right are synchronized, with some occasional phase slips. Meanwhile Up and Down have their own frequencies. In the second run, up and down move like in the first run. The right unit seems to have stopped oscillating. And the center site follows the oscillations of the left one. In the third run, up unit is synchronized with the left unit, though there is a big phase slip. The down and right units oscillate with independent frequencies. The center site evolves erratically. We can take away some conclusions. First, the experimental setup is heterogeneous in the actuators, which we already had noticed in other experiments. Second, achieving synchronization in the ferromagnet cross cell is more difficult than expected. The system is not very robust under disorder. Third, the system is chaotic. The three experimental runs do not show a unique steady state. This could be due to the way the experiment is initiated, which is poking the system with the hand. These leads to different initial conditions in the experimental

runs. Apparently, the system is sensitive to them.

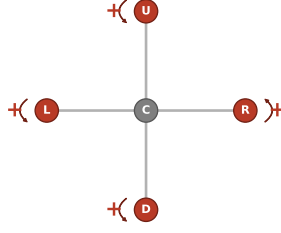


Figure 18: **Diagram of the ferromagnet cross cell.** The system is made up of four active units with the same chirality connected to a passive site in the center through viscoelastic bands. The cell is a small fragment of a ferromagnetic lattice.

In simulations, we observed that the system was chaotic. In the simple configuration of absolutely identical units, the system synchronizes. Moreover, the phenomenon is robust under small disorder in the parameters of the units. If one specific parameter of an individual unit is perturbed more significantly, for example by an order of 30% in activity, that particular unit desynchronizes from the rest. It can also happen that changing the parameter of one of the units also desynchronizes another unit. There is no special preference for which is the other unit that desynchronizes. It can be any of the other three active units. The choice is given by the unit whose parameters are less similar to the rest (to average of the rest). If the units are clustered in pairs in the parameter space, so that two units are similar to each other but quite different to the other two, then there are pairs of synchronization. Nevertheless, there is no overall synchronization. Interestingly, the passive unit synchronizes with the active units whose frequency is larger, although the detuning compared to the passive site's natural frequency (0 Hz) is larger than the detuning of the other active units. We think that the reason is that the frequency is determined by the activity, but the activity in turn also affects the amplitude. As we have seen in other sections, the amplitude has an effect on the Kuramoto coupling.

For all units to oscillate with different frequencies, one must make the different perturbations for every unit, which leads to huge differences in the parameters. The same example of perturbing the activity requires differences of more than 100% between some of the units. This shows that synchronization is indeed robust. One can always recover synchronization by increasing the neighbour coupling of the specific units that are greatly perturbed. This observation agrees with the conclusions derived from the Adler equation (6). Notice however, that the Adler equation only takes into account mismatches in the natural frequency. The simulations show that an analogous phenomenon happens for mismatches in other parameters, like activity or damping.

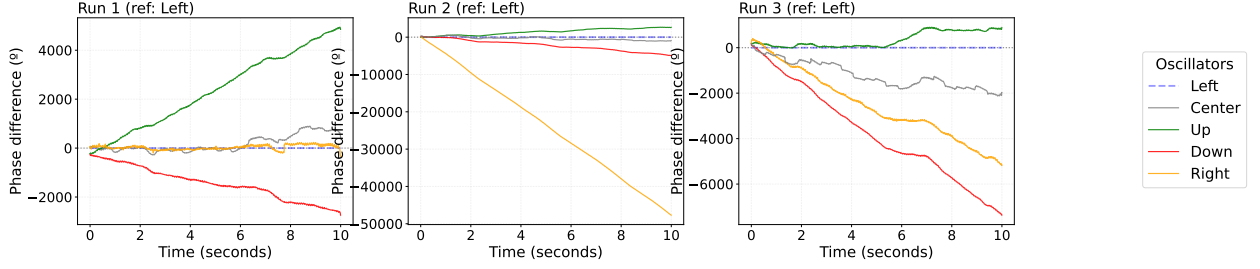


Figure 19: **Experimental data of ferromagnet cross cell.** The phase difference with respect to a reference (left unit) is plotted as time evolves for each unit. The different panels correspond to three experimental runs with the same parameter configuration.

6.2 Altermagnet cross cell

The altermagnet cross cell (figure 22) is made out of one passive unit in the center, two active units above (up) and below (down) the passive one with the same chirality, and two active units on the left and right of the passive one with opposite chirality.

Figure 23 shows the phase of each of the units in time, for different experimental runs. First, focus on the up and down units, which increase with a constant slope. Note that the down unit, depicted with orange line, is overlapped with the red line. This means that these two units are completely synchronized. This is the case in both experimental runs. Focusing on the first run, there are also two units, right and left, whose phase decreases in time. These units don't have the same phase, as we see that the lines separate with time. Hence, they are not synchronized. In comparison, on the plot on the right, the lines coincide for most of the experiment. At the end, it seems like they separate. The reason they separate is a phase slip. Phase slips are brief moments where the oscillators go out of sync before going back again into in-phase synchronization. Note that after the phase slip, the slope of both lines remains the same, meaning that their frequency is the same. So they synchronize again. The difference in experimental parameters between the two plots in 23 is the activity of the left and right units. For the plot on the left, $k_a^L = -4 = k_a^R$, while for the plot on the right $k_a^L = -7 = k_a^R$. Therefore, increasing the activity leads to better chances of synchronization. We already stated the same conclusion in the previous section.

There is a fifth unit, the passive one in the center, and it is actually the most interesting one in this system because it mediates the interactions between the active ones. Its phase exhibits step-like motion: there are straight lines and vertices where the slope of the straight line changes abruptly. An important feature is that the slope of the phase is either the same as the up and down units, or the same as the left and right units. Therefore, the center passive unit is sometimes synchronizing with the positive chirality active units and other times with the negative chirality units. It seems to be alternating between both scenarios. The system is therefore bistable, and it jumps from one stable state to the other. However, the unit is frustrated because it cannot synchronize with both

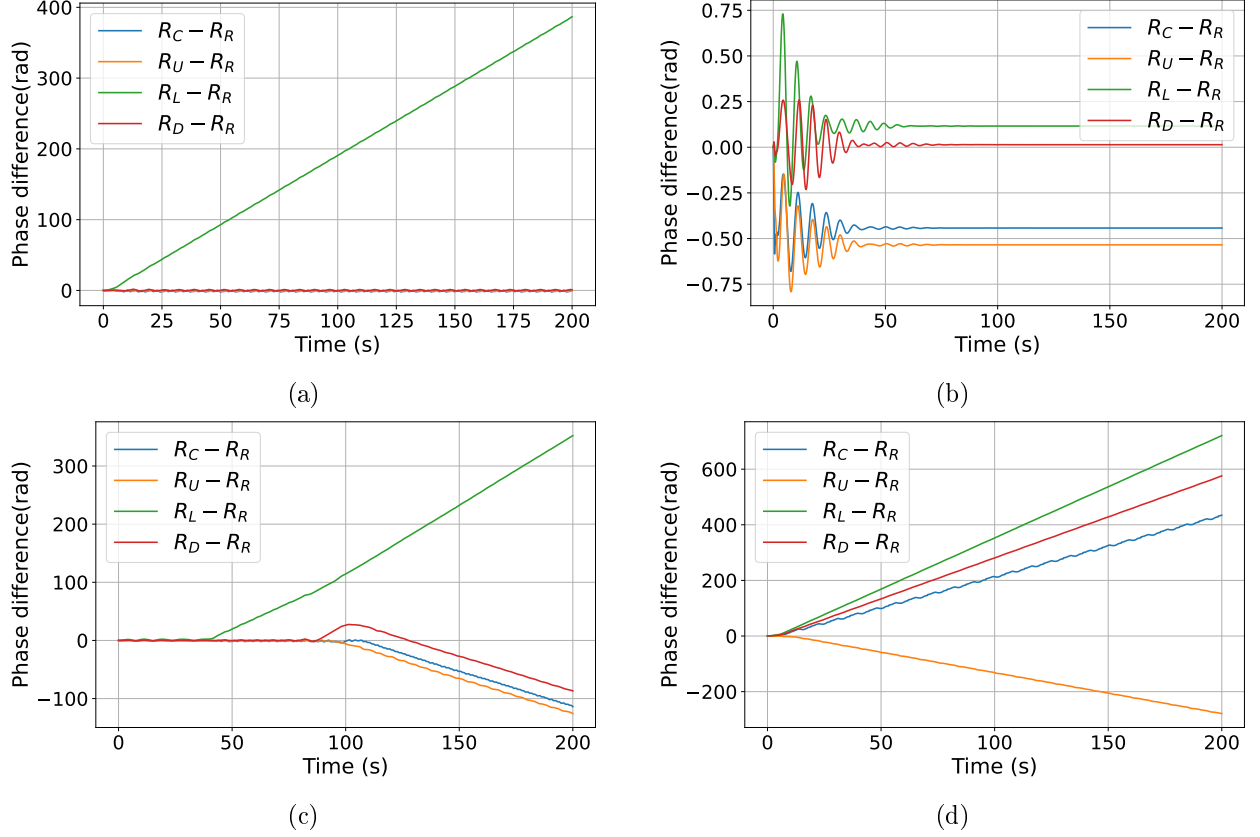


Figure 20: **Simulation data of the ferromagnet cross cell.** Each of the panels has a different behaviour. The parameters of the simulation are damping γ , onsite stiffness k_0 , nonlinear cubic saturation k_c , activity k_a and elastic coupling of the band k_b . The order in which the parameters of each unit are displayed is [Center, Up, Left, Down, Right]. **20a** The left unit is drifting with respect to the rest, which are synchronized. $\gamma = [1, 1.3, 1.17, 1.15, 1.13]$, $k_0 = [1.0, 2.0, 1.5, 2.0, 3.0]$, $k_c = 10.0$, $k_a = [0, 6, 10, 8, 7]$, $k_b = [6.0, 4.0, 5.0, 3.0]$. **20b** All the units synchronize. The only change compared to the parameters of 20a is the activity of the left unit $k_a = [0, 6, 10, 8, 7]$. **20c** The up, down and center units remain synchronized. The left and right units have their own frequency. The only parameter change with respect to 20a is the elastic neighbour band coupling of the left unit $k_b = [6.0, 5.0, 5.0, 3.0]$. Modifying the parameters of the left unit stops the synchronization of the right unit. **20d** No synchronization between any of the units. They all oscillate with different frequency. In order to achieve it, the disorder is increased. $\gamma = [1, 1.3, 1.7, 1.5, 1.3]$, $k_0 = [1.0, 2.0, 1.5, 2.0, 3.0]$, $k_c = 10.0$, $k_a = [0, 5, 15, 12, 7]$, $k_b = [6.0, 6.0, 5.0, 3.0]$.

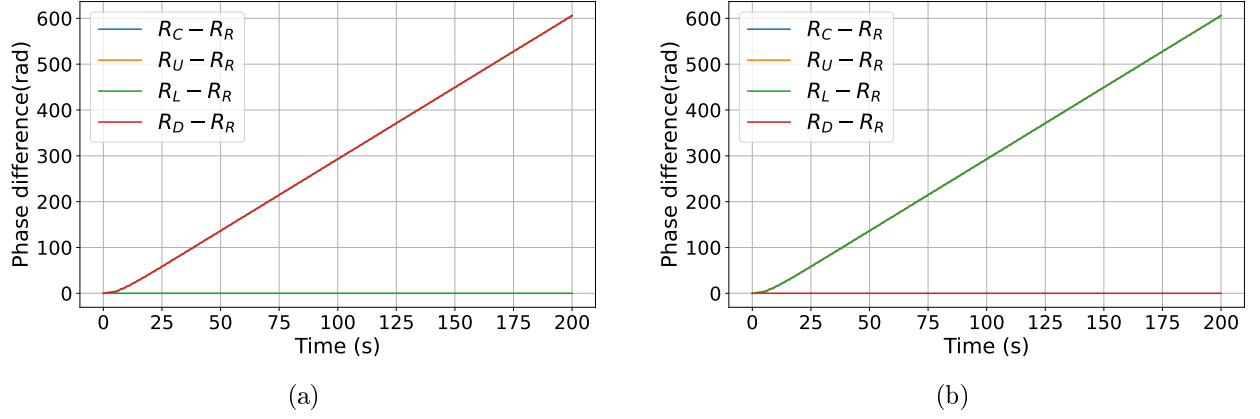


Figure 21: **Symmetric configurations in pairwise synchronization of ferromagnet cross cell.** The plots show the equivalence of the two possible pairwise synchronization configurations. **21a** Synchronization Up-Down and Left-Right. **21b** Synchronization Up-Left and Down-Right.

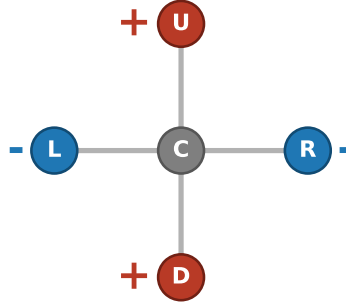


Figure 22: **Diagram of the altermagnet cross cell.** There are four active units connected to a passive central one through viscoelastic bands. Up and down units have positive chirality, while left and right units have negative chirality. The cell is a small fragment of the altermagnetic Lieb lattice.

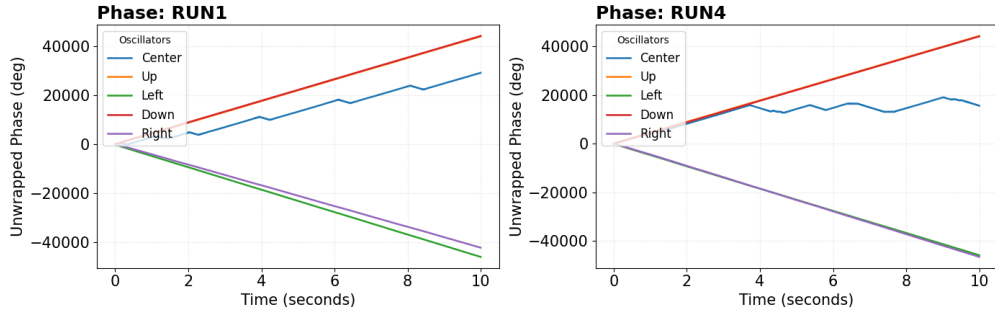


Figure 23: **Experimental data of altermagnet cross cell.** The phase of each unit is shown as time evolves. **Run 1:** The active units have the same absolute value of activity: $k_a^U = 4 = k_a^D$, $k_a^L = -4 = k_a^R$. Up and down units have their lines overlapped, which means they are synchronized. The left and right lines do not overlap, so they are not synchronized. This is because the passive site has a tendency to synchronize with the up and down units, since the slope is most of the time parallel to those units' lines. The passive site is responsible for the interactions leading to the synchronization of the positive-chirality units, and also for the drifting of the negative-chirality units. Such a tendency to synchronize with the up and down units is explained by the action of gravity in the vertical direction. Gravity pulls on the viscoelastic band, increasing its tension. This results in a slightly stronger coupling in the vertical direction compared to the horizontal direction. The band array is not fixed to the sides of the apparatus. **Run 4:** $k_a^U = 4 = k_a^D$, $k_a^L = -7 = k_a^R$. The activity of the negative-chirality units is increased in comparison with the previous experimental run. The notable effects are the synchronization of those units, the left and right ones. Moreover, the passive site's motion is more evenly distributed between the two stable states. Increasing the activity of the negative-chirality units seems to compensate for the asymmetry generated by gravity.

of the chiralities at the same time. One thing to keep in mind is that the passive unit is not tracing out a clear limit cycle like the active units. Its motion in phase space is more messy. Still, one can change from the displacement coordinates of both actuators to polar coordinates $(x_1, x_2) \rightarrow (R, \phi)$ and study its phase.

A priori, the center unit can choose any of the active units to synchronize itself with. Nevertheless, figure 23 and other experiments show a tendency to synchronize with the up and down units more often than with the left and right units. Here, we go back to the discussion we had in the previous section about horizontal and vertical couplings, and the slight difference in results between them. The general tendency of synchronizing with the up and down units is an indicator of the asymmetry between the vertical and horizontal coupling. From the results, one estimates that the vertical coupling, the one connecting the center node to up and down, is stronger than the horizontal coupling. The reason for the asymmetry is that gravity is acting on the vertical direction. Thus, the viscoelastic band is stretched and under more tension, making the effective elastic stiffness slightly stronger. Note that the viscoelastic band is fixed to the structure at the top, so that it doesn't fall, while it is not fixed to the sides. We think this is the reason for any asymmetry in the coupling directions and specifically, for the tendency of the center node to synchronize with its vertical neighbours.

We contrasted the experimental results with simulations of the altermagnet cross cell. For completely identical units, we obtained that the passive site alternates evenly between positive and negative chirality, as shown in figure 24a. In this case, the same chirality units are synchronized. So Up and Down units are synchronized to a positive frequency, while the Left and Right units are synchronized to a negative frequency. However, by introducing a small disorder in the damping, the active units of same chirality do not synchronize. The results are depicted in figure 24b. Moreover, the motion of the passive unit becomes very erratic. There is no noise added to the system, so the simulation is deterministic. The changes in the frequency of the passive unit are due to chaotic behaviour.

7 Bilayer Lieb lattice

In order to replace the spin degree of freedom at the sites of a Lieb lattice with mechanical oscillators, we need two Lieb-lattice layers of linear actuators. As before, pairs of actuators mirrored across the wall interact non-reciprocally. Within each Lieb-lattice layer, neighbouring actuators are coupled by viscoelastic bands.

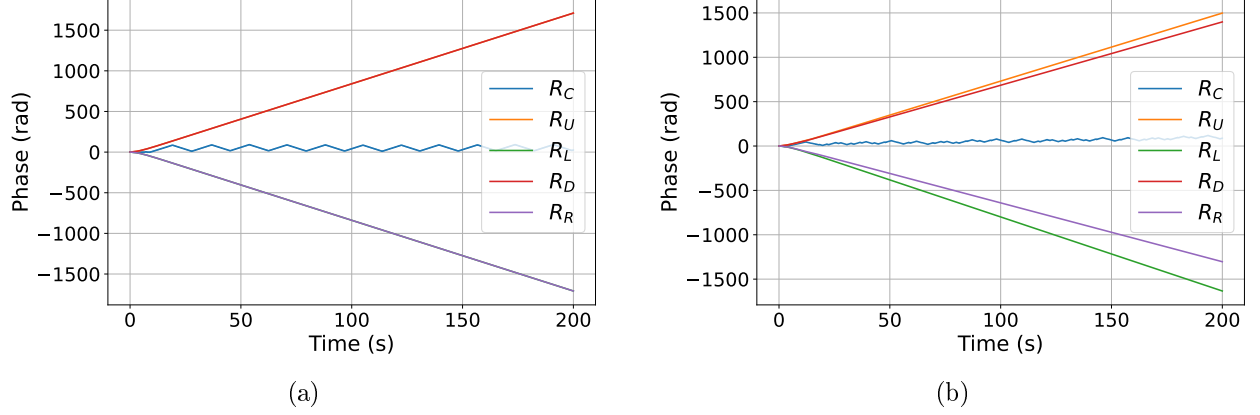


Figure 24: **Simulation data of electromagnetic cross cell, with and without disorder.** The phase of each unit is shown as time evolves. The parameters of the simulation: $k_0 = 2$ $k_c = 10$ $k_a = 10$ $k_b = 3$. **24a** Simulation with no disorder: $\gamma = 1$ for all units. Units with same chirality synchronize. The passive unit alternates evenly between synchronization with positive and negative chiralities, yielding a seesaw line. **24b** Simulation with quenched disorder in the damping: $\gamma = (1.1, 1.3, 1.2, 1.4, 1.5)$. Units with the same chirality do not synchronize. The passive unit alternates between bistable states chaotically.

7.1 Simulations of the overdamped regime

We wanted to observe if spin-momentum locking was present in the mechanical Lieb lattice. This meant observing positive chirality waves propagating in one direction, and negative chirality waves in another direction. In order to do so, we had to set the activity to be in the overdamped regime before the Hopf bifurcation. In this regime, friction dominates and waves are attenuated. This is a good thing to observe spin-momentum locking. The bad thing is that this regime is difficult to access experimentally. Since the waves are attenuated, the actuator displacements are very small and cannot be measured. For that reason we resorted to simulations.

The simulation consists of a 9×9 Lieb lattice of non-reciprocal units. The initial condition is a perturbation in the lower left corner of the lattice. We observe the waves that propagate through the system. Since friction dominates, they eventually die out. The motion of the active units is constrained by their built-in chirality. Nevertheless, we can analyse the chirality of the passive sites to know where each chirality propagated. We do so by measuring the angular momentum of the passive sites: $J = x_1 \dot{x}_2 - x_2 \dot{x}_1$. If $J > 0$, it means that the passive site has positive chirality, and if $J < 0$, negative chirality. For simplicity, we show in figure 25 the angular momentum of the passive sites on the edges of the Lieb lattice. We considered three initial conditions. The first one, figure 25a, has a kick as initial condition. The kick has no associated chirality, and so it is displayed in the figure as a 0. We observe that the chirality of the passive sites to the right is positive, while the chirality of the passive sites towards the top is negative. As in the case of a magnetic Lieb lattice, the positive chirality component of the perturbation is amplified in the horizontal direction. That is why we see the bottom and the right edges with a plus sign. In contrast, the negative component

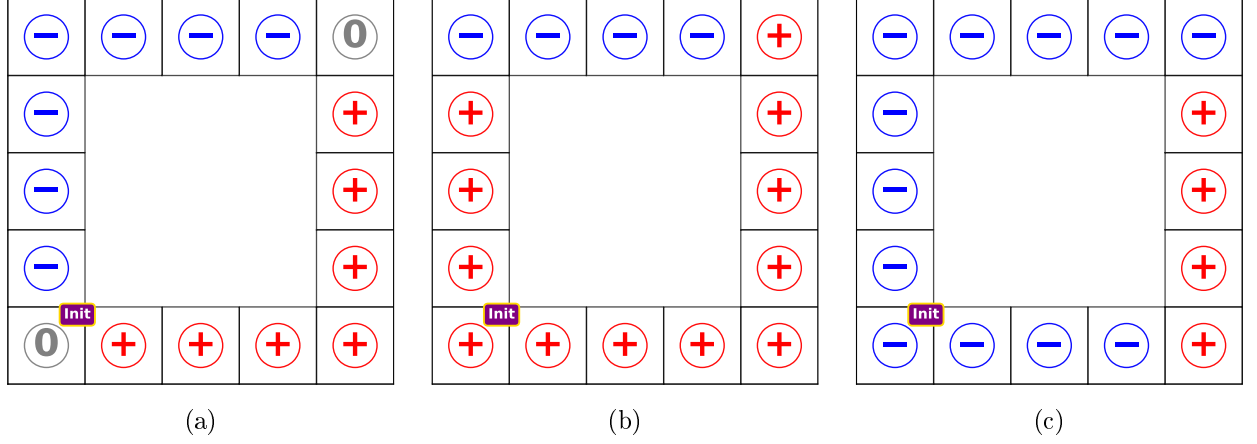


Figure 25: **Chirality-momentum locking in mechanical Lieb lattice.** Diagram illustrating the chirality of the passive sites in the edges of the mechanical Lieb lattice. It is a simulation of the overdamped regime. The chirality is measured as the average angular momentum of the unit: $J = x_1\dot{x}_2 - x_2\dot{x}_1$. The system is perturbed initially on the bottom left corner. We consider three different initial conditions, depending on the chirality of the perturbation. **25a** The initial perturbation is a kick with no chirality. The positive-chirality component is amplified in the horizontal direction. Thus, the bottom and right edges have positive chirality. The negative-chirality component is amplified in the vertical direction. Hence, the left and top edges have negative chirality. **25b** Although the initial perturbation has positive chirality, the top edge remains with negative chirality. **25c** Although the initial perturbation has negative chirality, the right edge remains with negative chirality. The property of chirality-momentum locking is independent of the initial perturbation.

is amplified vertically, and we observe the left and top edges with a minus sign. This is precisely the same phenomenon of spin-momentum locking observed in altermagnets.

In subfigure 25b, the initial condition is changed to a positive chirality perturbation. The difference is on the left edge, which is now positive due to the initial condition. Nevertheless, we still observe that the top edge has negative chirality, even though the perturbation is of positive chirality. A similar thing happens in the last panel 25c, where the initial condition is a negative chirality perturbation. Although the closest passive sites acquire negative chirality, the right edge still displays positive chirality.

8 Conclusion

In conclusion, we implemented oscillator chirality in a metamaterial platform. The original idea behind the implementation was to replace the spin angular momentum of magnetic materials. Chirality was achieved by coupling two actuators non-reciprocally. The actuators performed limit cycles in phase space that behaved like an oscillator. It allowed the study of systems of various dimensions and symmetries. Some of those were analogous to ferromagnets, antiferromagnets and altermagnets, but in a classical setting. Our main motivation was the study of recently discovered altermagnets,

which have become a trending topic in condensed matter physics. The motivation made us switch disciplines quickly to the fields of nonlinear dynamics, metamaterials and soft matter. The experimental milestone was the construction of a Lieb lattice of mechanical actuators. We started by analysing its microscopic properties, which led us to the theory of nonlinear oscillators. After that, we analyse the macroscopic behaviour of the system, and also explored different geometries and symmetries. The analysis included experimental results, but also comparison with theoretical predictions and further investigations using simulations.

Despite the physical actuators being imperfect or defective, we still observed signs of the corresponding scalings predicted by the supercritical Hopf bifurcation. This result was a nice sanity check. There were two different regimes, either before or after the Hopf bifurcation. Before the bifurcation, friction dominated. The oscillations were attenuated and the system came to rest. By increasing the activity parameter, i.e. the energy input, the system underwent the Hopf bifurcation, resulting in the formation of limit cycles. The clockwise direction of the limit cycle was associated with negative chirality, and the counterclockwise direction with positive chirality. There was a third type of unit without activity, called passive. We studied both regimes separated by the bifurcation. However, it was difficult to measure experimentally in the overdamped regime. In that case, we carried out simulations.

The next step was the study of pairwise interactions between the oscillatory units. We realised that units with the same chirality have a parameter region where they synchronize, similarly to other oscillatory systems. However, if the units had opposite chirality, we didn't observe synchronization. If passive units were added, they would absorb some energy from the active ones and use it to follow the motion of their neighbours. The theoretical models predict that synchronization is caused by the viscous interaction rather than the elastic one. This was an interesting result, because the primitive models did not include the viscous term.

Next we studied the one-dimensional chain of units. If the active units had the same chirality, they synchronized with a large amplitude in the regime after the Hopf bifurcation. In the regime below the bifurcation, perturbations propagated through the chain with small attenuation. If some active unit had opposite chirality, then the perturbations stopped propagating when reaching that particular unit. In the limit cycle regime, the oscillation amplitude was significantly reduced. These effects were an expression of selective propagation depending on the chirality of the units. We studied more complicated features like the transient dynamics and the effect of the system size using simulations. The transient dynamics, that takes the chain from the initial condition to the complete synchronization, is governed by wave coarsening. In experiments, synchronization happens very fast, indicating a very small synchronization timescale. The system size affects such timescale. For longer chains, one needs to wait more time to reach the synchronized steady state.

The experimental results up to now suggested that a ferromagnetic lattice of oscillators after the Hopf bifurcation would have all the units synchronized to a single frequency. For the antiferromagnetic

Lieb lattice, all the positive chirality units would synchronize together to some frequency, and all the negative chirality units would synchronize to another frequency. However, that is not what we observed when studying small fragments, the cross cells, of those systems. The synchronization phenomenon in the ferromagnet cross cell seems to be less robust than expected. The consequence is the experimental absence of synchronization. Simulations with small quenched disorder support the claim. The altermagnet cross cell is bistable, and its passive unit is frustrated between synchronizing with the positive and negative chirality units. The passive unit alternates chaotically between both states, even in simulations.

We simulated the mechanical Lieb lattice in the overdamped regime. The system exhibited chirality-momentum locking, like the original magnetic Lieb lattice. This allows the transmission of different signals by means of mechanical waves, in perpendicular directions. The signals are filtered nicely so there is no mixing. Nevertheless, these waves are significantly attenuated. To overcome this obstacle one would need to decrease the friction of the passive units. The friction of the active units can be compensated with activity.

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Appendices

A Calibration of actuators

We calibrated the properties of the linear actuators with two methods. The first involved poking them and measuring the damped oscillations. We fitted the position data to an exponentially attenuated wave: $f = Ae^{-\gamma t} \cos(\omega t + \phi) + c$. The results of the fit are shown in table 1. The second method was measuring the force-displacement curve on an Instron machine. The curves were measured for a single actuator at different rates. The results are shown in figure 26.

Node	Fit 1			Fit 2		
	Amplitude	Frequency	Damping	Amplitude	Frequency	Damping
1	1.0956e+00	2.0110e+02	3.9345e-03	4.0394e+00	2.0102e+02	3.6814e-03
2	-5.8440e-01	2.0102e+02	1.5981e-03	4.0625e+00	2.0102e+02	2.7045e-03
3	6.2493e-01	2.0102e+02	2.7074e-03	4.6398e-01	2.0110e+02	5.3336e-04

Table 1: **Fit parameters for calibration of three actuators.** The three parameters are amplitude, frequency, and damping. Two fits were done for each actuator.

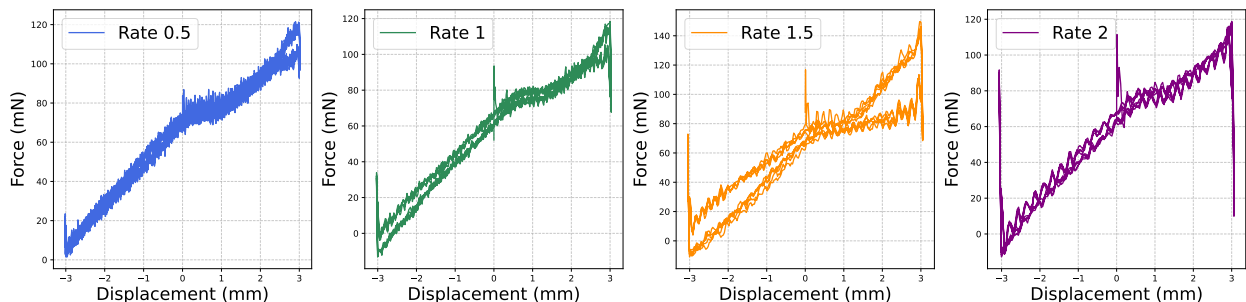


Figure 26: **Force-displacement curves of one actuator at different speeds.**

B Multiple scale expansion derivation

In this section we analyse the classical dynamical equations of two non-reciprocal oscillators (units) coupled together via a viscoelastic band. Each unit contains two actuators. The configuration is displayed in figure 9. We use the method of multiple scales to find the complex amplitude equations describing the slow dynamics of the system. Next we switch to polar variables: real amplitude and phase. Finally, we interpret the results in terms of synchronization between the two oscillators.

Since there are 2 actuators per non-reciprocal unit, we have 4 actuators and 8 degrees of freedom in total. We denote the displacements of the actuators in unit 1 as $x_{1\alpha}$ and $x_{1\beta}$, and in unit 2 as

$x_{2\alpha}$ and $x_{2\beta}$. The actuators follow classical mechanics laws, so we can use Newton's second law to write down the equations of motion:

$$\begin{aligned}
\ddot{x}_{1\alpha} + \omega^2 x_{1\alpha} &= \epsilon(\omega^2 - \omega_{1\alpha}^2)x_{1\alpha} - \epsilon\gamma_{1\alpha}\dot{x}_{1\alpha} - \epsilon k_{c,1\alpha}x_{1\alpha}^3 + \epsilon k_a^1 x_{1\beta} + \epsilon k_v(\dot{x}_{2\alpha} - \dot{x}_{1\alpha}) + \frac{\epsilon k_b}{2l^2}(x_{2\alpha} - x_{1\alpha})^3, \\
\ddot{x}_{1\beta} + \omega^2 x_{1\beta} &= \epsilon(\omega^2 - \omega_{1\beta}^2)x_{1\beta} - \epsilon\gamma_{1\beta}\dot{x}_{1\beta} - \epsilon k_{c,1\beta}x_{1\beta}^3 - \epsilon k_a^1 x_{1\alpha} + \epsilon k_v(\dot{x}_{2\beta} - \dot{x}_{1\beta}) + \frac{\epsilon k_b}{2l^2}(x_{2\beta} - x_{1\beta})^3, \\
\ddot{x}_{2\alpha} + \omega^2 x_{2\alpha} &= \epsilon(\omega^2 - \omega_{2\alpha}^2)x_{2\alpha} - \epsilon\gamma_{2\alpha}\dot{x}_{2\alpha} - \epsilon k_{c,2\alpha}x_{2\alpha}^3 + \epsilon k_a^2 x_{2\beta} + \epsilon k_v(\dot{x}_{1\alpha} - \dot{x}_{2\alpha}) + \frac{\epsilon k_b}{2l^2}(x_{1\alpha} - x_{2\alpha})^3, \\
\ddot{x}_{2\beta} + \omega^2 x_{2\beta} &= \epsilon(\omega^2 - \omega_{2\beta}^2)x_{2\beta} - \epsilon\gamma_{2\beta}\dot{x}_{2\beta} - \epsilon k_{c,2\beta}x_{2\beta}^3 - \epsilon k_a^2 x_{2\alpha} + \epsilon k_v(\dot{x}_{1\beta} - \dot{x}_{2\beta}) + \frac{\epsilon k_b}{2l^2}(x_{1\beta} - x_{2\beta})^3,
\end{aligned} \tag{59}$$

where $\omega_{u\sigma}$ is the natural frequency of actuator $\sigma \in \{\alpha, \beta\}$ in unit $u \in \{1, 2\}$, $\gamma_{u\sigma}$ is the damping coefficient, $k_{c,u\sigma}$ is the nonlinear saturation stiffness, k_a^u is the non-reciprocal coupling within unit u , k_b and k_v are the elastic and viscous coupling strengths of the viscoelastic band, and l is the distance between the actuators at rest. The parameter ϵ indicates that all the perturbations, detuning, damping, nonlinear stiffness, non-reciprocal coupling and neighbour couplings, are weak compared to the harmonic oscillator terms.

The first step is to perform the multiple scale expansion. We skip some of the details as we already presented them before. The result is a set of 4 coupled complex amplitude equations describing the slow dynamics of the system. The solution at order ϵ^0 is

$$x_{u\sigma}^0 = A_{u\sigma}(T)e^{i\omega t} + A_{u\sigma}^*(T)e^{-i\omega t}, \tag{60}$$

for all actuators ($u = 1, 2$, $\sigma \in \{\alpha, \beta\}$). When plugging the above formula in the cubic term of the $\mathcal{O}(\epsilon)$ equation, one gets

$$(x_{u\sigma}^0)^3 = A_{u\sigma}^3 e^{3i\omega t} + 3 \underbrace{A_{u\sigma}^2 A_{u\sigma}^*}_{|A_{u\sigma}|^2 A_{u\sigma}} e^{i\omega t} + 3 \underbrace{|A_{u\sigma}|^2 A_{u\sigma}^* e^{-i\omega t} + (A_{u\sigma}^*)^3 e^{-3i\omega t}}_{+ \text{ c.c.}}. \tag{61}$$

For the neighbour coupling term between the α -actuators in the two units, we do a similar procedure, but now we have two different oscillators. The full term is

$$(x_{2\alpha} - x_{1\alpha})^3 = x_{2\alpha}^3 - 3x_{2\alpha}^2 x_{1\alpha} + 3x_{2\alpha} x_{1\alpha}^2 - x_{1\alpha}^3. \tag{62}$$

Before raising to the third power, we can substitute the $\mathcal{O}(\epsilon^0)$ solution for $x_{2\alpha}$ and $x_{1\alpha}$:

$$x_{2\alpha}^0 - x_{1\alpha}^0 = \underbrace{(A_{2\alpha} - A_{1\alpha})}_B e^{i\omega t} + \underbrace{(A_{2\alpha}^* - A_{1\alpha}^*)}_{B^*} e^{-i\omega t} = \Delta. \tag{63}$$

Raising to the cube, we find terms with frequencies 3ω , -3ω , ω and $-\omega$. The terms with frequencies 3ω and -3ω disappear when applying the solvability condition, since they are the only terms with

those frequencies, and they must be set to zero. We can ignore them. The only term that survives is the one with frequency ω , which is given by

$$\Delta^3 = \underbrace{B^3 e^{3i\omega t}}_{\text{vanish}} + 3|B|^2 B e^{i\omega t} + 3|B|^2 B^* e^{-i\omega t} + \underbrace{B^{*3} e^{-3i\omega t}}_{\text{vanish}} = 3|A_{2\alpha} - A_{1\alpha}|^2 (A_{2\alpha} - A_{1\alpha}) e^{i\omega t} + \text{c.c.} \quad (64)$$

Recall the identities from the multiple scale expansion:

$$x_{u\sigma} = x_{u\sigma}^0 + \epsilon x_{u\sigma}^1 \quad \text{and} \quad \frac{d}{dt} = \partial_t + \epsilon \partial_T. \quad (65)$$

Substituting the above identities in the equations of motion, and looking at order $\mathcal{O}(\epsilon^1)$, we get the following equation for the first actuator in unit 1:

$$\begin{aligned} \epsilon^1 : \quad & \partial_{tt} x_{1\alpha}^1 - 2\partial_{tT} x_{1\alpha}^0 + \omega^2 x_{1\alpha}^1 = \\ & (\omega^2 - \omega_{1\alpha}^2) x_{1\alpha}^0 - \gamma_{1\alpha} \dot{x}_{1\alpha}^0 - k_{c,1\alpha} (x_{1\alpha}^0)^3 + k_a^1 x_{1\beta}^0 + k_v (\dot{x}_{2\alpha}^0 - \dot{x}_{1\alpha}^0) + \frac{k_b}{2l^2} (x_{2\alpha}^0 - x_{1\alpha}^0)^3. \end{aligned} \quad (66)$$

Let's look at mode $e^{i\omega t}$ and apply the solvability condition:

$$\begin{aligned} & (\omega^2 - \omega_{1\alpha}^2) A_{1\alpha} - i\omega \gamma_{1\alpha} A_{1\alpha} - 2i\omega A_{1\alpha}' - 3k_{c,1\alpha} |A_{1\alpha}|^2 A_{1\alpha} \\ & + k_a^1 A_{1\beta} + i\omega k_v (A_{2\alpha} - A_{1\alpha}) + \frac{k_b}{2l^2} 3|A_{2\alpha} - A_{1\alpha}|^2 (A_{2\alpha} - A_{1\alpha}) = 0. \end{aligned} \quad (67)$$

Reorganizing the expression we get the Schrödinger-like equation:

$$\begin{aligned} iA_{1\alpha}' = \frac{1}{2\omega} \left[\underbrace{(\omega^2 - \omega_{1\alpha}^2)}_{(\omega - \omega_{1\alpha})(\omega + \omega_{1\alpha}) \approx 2\omega(\omega - \omega_{1\alpha})} A_{1\alpha} - i\omega \gamma_{1\alpha} A_{1\alpha} - 3k_{c,1\alpha} |A_{1\alpha}|^2 A_{1\alpha} + k_a^1 A_{1\beta} \right. \\ \left. + \frac{ik_v}{2} (A_{2\alpha} - A_{1\alpha}) + \frac{3k_b}{2l^2} |A_{2\alpha} - A_{1\alpha}|^2 (A_{2\alpha} - A_{1\alpha}) \right]. \end{aligned} \quad (68)$$

After simplifying the expression, we get the complex amplitude equation for the first actuator in unit 1:

$$A_{1\alpha}' = -i(\omega - \omega_{1\alpha}) A_{1\alpha} - \frac{1}{2} \gamma_{1\alpha} A_{1\alpha} + \frac{3ik_{c,1\alpha}}{2\omega} |A_{1\alpha}|^2 A_{1\alpha} \quad (69)$$

$$+ \frac{3ik_a^1}{2\omega} A_{1\beta} + \frac{k_v}{2} (A_{2\alpha} - A_{1\alpha}) - \frac{3ik_b}{4\omega l^2} |A_{2\alpha} - A_{1\alpha}|^2 (A_{2\alpha} - A_{1\alpha}) \quad (70)$$

The next step is to rewrite everything in terms of the polar variables $A_{u\sigma} = R_{u\sigma}(T) e^{i\phi_{u\sigma}(T)}$ and simplify:

$$|A_{1\alpha}|^2 = R_{1\alpha}^2 \quad |A_{2\alpha} - A_{1\alpha}|^2 = \left| R_{2\alpha}e^{i\phi_{2\alpha}} - R_{1\alpha}e^{i\phi_{1\alpha}} \right|^2 \quad (71)$$

The second term can be worked out:

$$\left(R_{2\alpha}e^{i\phi_{2\alpha}} - R_{1\alpha}e^{i\phi_{1\alpha}} \right) \left(R_{2\alpha}e^{-i\phi_{2\alpha}} - R_{1\alpha}e^{-i\phi_{1\alpha}} \right) = R_{2\alpha}^2 + R_{1\alpha}^2 - 2R_{2\alpha}R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha}) \quad (72)$$

Multiplying the previous expression by $(A_{2\alpha} - A_{1\alpha})$:

$$\begin{aligned} \cdot (A_{2\alpha} - A_{1\alpha}) &\rightarrow (R_{2\alpha}^2 + R_{1\alpha}^2 - 2R_{2\alpha}R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha})) \left(R_{2\alpha}e^{i\phi_{2\alpha}} - R_{1\alpha}e^{i\phi_{1\alpha}} \right) = \\ &= R_{2\alpha}^3e^{i\phi_{2\alpha}} + R_{1\alpha}^2R_{2\alpha}e^{i\phi_{2\alpha}} - 2R_{2\alpha}^2R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha})e^{i\phi_{2\alpha}} \\ &\quad - R_{2\alpha}^2R_{1\alpha}e^{i\phi_{1\alpha}} - R_{1\alpha}^3e^{i\phi_{1\alpha}} + 2R_{2\alpha}R_{1\alpha}^2\cos(\phi_{2\alpha} - \phi_{1\alpha})e^{i\phi_{1\alpha}} \end{aligned} \quad (73)$$

We separate the real part, which will be the one associated to the radius/amplitude:

$$\begin{aligned} \text{Re} [|A_{2\alpha} - A_{1\alpha}|^2 (A_{2\alpha} - A_{1\alpha})] &= R_{2\alpha}^3 \cos \phi_{2\alpha} + R_{1\alpha}^2 R_{2\alpha} \cos \phi_{2\alpha} \\ &\quad - 2R_{2\alpha}^2 R_{1\alpha} \cos(\phi_{2\alpha} - \phi_{1\alpha}) \cos \phi_{2\alpha} \\ &\quad - R_{2\alpha}^2 R_{1\alpha} \cos \phi_{1\alpha} - R_{1\alpha}^3 \cos \phi_{1\alpha} \\ &\quad + 2R_{2\alpha} R_{1\alpha}^2 \cos(\phi_{2\alpha} - \phi_{1\alpha}) \cos \phi_{1\alpha} \end{aligned} \quad (74)$$

The imaginary part is very similar with $\sin \phi_{1\alpha}$ or $\sin \phi_{2\alpha}$ terms instead of cosines.

Putting everything back into equation (46) after substituting the polar variables, we find:

$$\begin{aligned} \dot{R}_{1\alpha}e^{i\phi_{1\alpha}} + iR_{1\alpha}\dot{\phi}_{1\alpha}e^{i\phi_{1\alpha}} &= -i(\omega - \omega_{1\alpha})R_{1\alpha}e^{i\phi_{1\alpha}} - \frac{1}{2}\gamma_{1\alpha}R_{1\alpha}e^{i\phi_{1\alpha}} + \frac{3ik_{c,1\alpha}}{2\omega}R_{1\alpha}^3e^{i\phi_{1\alpha}} - \frac{3ik_a^1}{2\omega}R_{1\beta}e^{i\phi_{1\beta}} \\ &\quad + \frac{k_v}{2}(R_{2\alpha}e^{i\phi_{2\alpha}} - R_{1\alpha}e^{i\phi_{1\alpha}}) - \frac{3ik_b}{4\omega l^2}[\text{equation (73)}] \end{aligned} \quad (75)$$

We can multiply both sides by $e^{-i\phi_{1\alpha}}$.

$$\begin{aligned} \dot{R}_{1\alpha} + iR_{1\alpha}\dot{\phi}_{1\alpha} &= -i(\omega - \omega_{1\alpha})R_{1\alpha} - \frac{1}{2}\gamma_{1\alpha}R_{1\alpha} + \frac{3ik_{c,1\alpha}}{2\omega}R_{1\alpha}^3 - \frac{3ik_a^1}{2\omega}R_{1\beta}e^{i(\phi_{1\beta} - \phi_{1\alpha})} \\ &\quad + \frac{k_v}{2}(R_{2\alpha}e^{i(\phi_{2\alpha} - \phi_{1\alpha})} - R_{1\alpha}) - \frac{3ik_b}{4\omega l^2} \left[e^{i(\phi_{2\alpha} - \phi_{1\alpha})} (R_{2\alpha}^3 + R_{1\alpha}^2 R_{2\alpha} - 2R_{2\alpha}^2 R_{1\alpha} \cos(\phi_{2\alpha} - \phi_{1\alpha})) \right. \\ &\quad \left. - R_{2\alpha}^2 R_{1\alpha} - R_{1\alpha}^3 + 2R_{2\alpha} R_{1\alpha}^2 \cos(\phi_{2\alpha} - \phi_{1\alpha}) \right] \end{aligned} \quad (76)$$

We observe that the real part is associated to the radius/amplitude dynamics, while the imaginary part is associated to the phase dynamics. We take the real and imaginary parts of the previous expression. The real part is:

$$\begin{aligned}\dot{R}_{1\alpha} = & -\frac{1}{2}\gamma_{1\alpha}R_{1\alpha} + \frac{3k_a^1}{2\omega}R_{1\beta}\sin(\phi_{1\beta} - \phi_{1\alpha}) + \frac{k_v}{2}(R_{2\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha}) - R_{1\alpha}) \\ & + \frac{3k_b}{4\omega l^2}\sin(\phi_{2\alpha} - \phi_{1\alpha})(R_{2\alpha}^3 + R_{1\alpha}^2R_{2\alpha} - 2R_{2\alpha}^2R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha}))\end{aligned}\quad (77)$$

The imaginary part of the equation is:

$$\begin{aligned}R_{1\alpha}\dot{\phi}_{1\alpha} = & -(\omega - \omega_{1\alpha})R_{1\alpha} + \frac{3k_{c,1\alpha}}{2\omega}R_{1\alpha}^3 - \frac{3k_a^1}{2\omega}R_{1\beta}\cos(\phi_{1\beta} - \phi_{1\alpha}) + \frac{k_v}{2}(R_{2\alpha}\sin(\phi_{2\alpha} - \phi_{1\alpha}) - R_{1\alpha}) \\ & - \frac{3k_b}{4\omega l^2}[\cos(\phi_{2\alpha} - \phi_{1\alpha})(R_{2\alpha}^3 + R_{1\alpha}^2R_{2\alpha} - 2R_{2\alpha}^2R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha})) \\ & - R_{2\alpha}^2R_{1\alpha} - R_{1\alpha}^3 + 2R_{2\alpha}R_{1\alpha}^2\cos(\phi_{2\alpha} - \phi_{1\alpha})]\end{aligned}\quad (78)$$

Dividing the imaginary part by $R_{1\alpha}$ yields the phase dynamics for $\dot{\phi}_{1\alpha}$:

$$\begin{aligned}\dot{\phi}_{1\alpha} = & -(\omega - \omega_{1\alpha}) + \frac{3k_{c,1\alpha}}{2\omega}R_{1\alpha}^2 - \frac{3k_a^1}{2\omega}\frac{R_{1\beta}}{R_{1\alpha}}\cos(\phi_{1\beta} - \phi_{1\alpha}) + \frac{k_v}{2}\left(\frac{R_{2\alpha}}{R_{1\alpha}}\sin(\phi_{2\alpha} - \phi_{1\alpha}) - 1\right) \\ & - \frac{3k_b}{4\omega l^2}\left[\cos(\phi_{2\alpha} - \phi_{1\alpha})\left(\frac{R_{2\alpha}^3}{R_{1\alpha}} + R_{1\alpha}R_{2\alpha} - 2R_{2\alpha}^2\cos(\phi_{2\alpha} - \phi_{1\alpha})\right) \right. \\ & \left. - R_{2\alpha}^2 - R_{1\alpha}^2 + 2R_{2\alpha}R_{1\alpha}\cos(\phi_{2\alpha} - \phi_{1\alpha})\right]\end{aligned}\quad (79)$$

The equations for the other actuators (1β , 2α , 2β) are obtained in the same way and they will just differ in some of the indices. We have arrived at the envelope/Stuart-Landau equations of two non-reciprocal units coupled via a viscoelastic band. We can compare the coefficients of (47) with experimental data. Note that equation (47) resembles a Kuramoto-type of model. The first term is the natural frequency of the spring. The second term is a non-isochronous term. It means that how large the radius is affects the frequency of oscillation. The third term is a usual Kuramoto coupling between two oscillators. The important feature here is that the coupling is proportional to the ratio of radii $\frac{R_{2\alpha}}{R_{1\alpha}}$. The last term is a more complicated Kuramoto-type of coupling, which is a consequence of the cubic neighbour coupling.

We can also consider some simplified limits of equations (46) and (47). Actuators α and β in each unit are coupled non-reciprocally, so their phase difference is $|\phi_{1\beta} - \phi_{1\alpha}| = \pi/2$ and $|\phi_{2\beta} - \phi_{2\alpha}| = \pi/2$. This means that the non-reciprocal coupling term is maximal in the radius equation, and it is zero

in the phase equation. We can also consider the synchronization regime, when the limit cycles of units 1 and 2 oscillate with same frequency. In the case of complete synchronization between the α -actuators, both have the same phase: $\phi_{2\alpha} = \phi_{1\alpha}$. In this limit, the neighbour coupling term in the amplitude equation vanishes, and the neighbour coupling term simplifies.

$$\dot{R}_{1\alpha} = -\frac{1}{2}\gamma_{1\alpha}R_{1\alpha} + \frac{3k_a^1}{2\omega}R_{1\beta} + \frac{k_v}{2}(R_{2\alpha} - R_{1\alpha}) \quad (80)$$

$$\dot{\phi}_{1\alpha} = -(\omega - \omega_{1\alpha}) + \frac{3k_{c,1\alpha}}{2\omega}R_{1\alpha}^2 - \frac{k_v}{2} + \frac{3k_b}{4\omega l^2} \frac{(R_{2\alpha} - R_{1\alpha})^2}{R_{1\alpha}} \quad (81)$$

We can take the phase dynamics approximation, and to be valid it requires the RHS of equation (48) to be zero. That gives a constraint on the radii of the actuators and the damping coefficient: $R_{1\beta} = \frac{\omega\gamma_{1\alpha}}{3k_a^1}R_{1\alpha}$. By considering also the $\dot{R}_{1\beta}$ equation and the same simplifying assumptions, we obtain a relation between the radii of the actuators inside the unit and the damping coefficient:

$$\frac{\gamma_{1\alpha}}{\gamma_{1\beta}} = \left(\frac{R_{1\beta}}{R_{1\alpha}} \right)^2 \quad (82)$$

The equation seems reasonable. The higher the damping for one of the springs inside a non-reciprocal unit, the smaller its amplitude of oscillation.

If the neighbour coupling was linear instead of cubic, we would have obtained

$$k_b(A_{2\alpha} - A_{1\alpha}) = k_b(R_{2\alpha}e^{i\phi_{2\alpha}} - R_{1\alpha}e^{i\phi_{1\alpha}}) \longrightarrow k_b \left(1 - \frac{R_{2\alpha}}{R_{1\alpha}} \cos(\phi_{2\alpha} - \phi_{1\alpha}) \right). \quad (83)$$

If we view this term as a Kuramoto-type of coupling, we see that the coupling constant is given by $k_b R_{2\alpha}/R_{1\alpha}$. When doing experiments at first, the individual amplitude of oscillations of 2 *uncoupled* non-reciprocal units was different. And when they were coupled with the viscoelastic band, the amplitudes changed in a nontrivial way. I suspected that the amplitude had an effect on the Kuramoto coupling. My guess was that if we have the Kuramoto system

$$\dot{\Theta}_1 = \omega_1 + K_1 \sin(\Theta_2 - \Theta_1) \dot{\Theta}_2 = \omega_2 + K_2 \sin(\Theta_1 - \Theta_2), \quad (84)$$

then the coupling constants were proportional to the other oscillator's amplitude: $K_1 \propto R_{2\alpha}$ and $K_2 \propto R_{1\alpha}$. This ansatz explained the qualitative behaviour of the synchronization frequency and amplitude of the coupled system. However, it didn't explain the quantitative results. One reason that could explain this disagreement is the fact that the coupling is not linear, but cubic. The neighbour coupling term is given by $|A_{2\alpha} - A_{1\alpha}|^2(A_{2\alpha} - A_{1\alpha})$, which is a more complicated function of the amplitudes and phases of the oscillators.

Assuming that both units are tracing out stable limit cycles, so the phase differences are $\phi_{1\beta} - \phi_{1\alpha} = \pi/2 = \phi_{2\beta} - \phi_{2\alpha}$ and the radii are fixed $\dot{R}_{u\sigma} = 0$, the whole dynamical system reduces to

$$\dot{\phi}_{1\alpha} = \omega_{1\alpha} + \frac{3k_{c,1\alpha}}{2\omega} R_{1\alpha}^2 + \frac{k_v}{2} \left(\frac{R_{2\alpha}}{R_{1\alpha}} \sin(\phi_{2\alpha} - \phi_{1\alpha}) - 1 \right) + [k_b \text{ terms}] \quad (85)$$

$$\dot{\phi}_{2\alpha} = \omega_{2\alpha} + \frac{3k_{c,2\alpha}}{2\omega} R_{2\alpha}^2 + \frac{k_v}{2} \left(\frac{R_{1\alpha}}{R_{2\alpha}} \sin(\phi_{1\alpha} - \phi_{2\alpha}) - 1 \right) + [k_b \text{ terms}]. \quad (86)$$

We are interested in the phase difference between the two units $\Delta\phi = \phi_{2\alpha} - \phi_{1\alpha}$.

The dynamics of the phase difference is:

$$\dot{\Delta\phi} = \omega_{2\alpha} - \omega_{1\alpha} + \frac{3}{2\omega} (k_{c,2\alpha} R_{2\alpha}^2 - k_{c,1\alpha} R_{1\alpha}^2) - \frac{k_v}{2} \sin \Delta\phi \left(\frac{R_{2\alpha}}{R_{1\alpha}} + \frac{R_{1\alpha}}{R_{2\alpha}} \right) + [k_b \text{ terms}] \quad (87)$$

To simplify further, we consider identical oscillators $R_{1\alpha} = R_{2\alpha}$, $k_{c,1\alpha} = k_{c,2\alpha}$. Note that the k_b terms are even due to the cosine function. Since we subtract two identical even functions (with opposite argument $-\Delta\phi$), they cancel out. The expression nicely simplifies to

$$\dot{\Delta\phi} = \underbrace{\omega_{2\alpha} - \omega_{1\alpha}}_{\text{detuning}} - k_v \sin \Delta\phi, \quad (88)$$

which is precisely the Adler equation. The assumptions made were: identical units, quadrature phase within each unit, and stable limit cycles of individual units.

C Two units

In figure 27, we show the experimental data of the phase difference between two units with the same chirality. We can clearly see constant lines that correspond to phase locking. There are also occasional phase slips. The figure justifies studying the frequency changes from uncoupled to coupled units.

D 1D chain

We can also compare how the limit cycles look in both scenarios: with same and with opposite chirality (see figure 28). The results from the same chirality (figure 28a) show circular thin limit cycles. The fact that they are circular means that the phase difference between the two actuators is 90° . The fact that they are thin is a consequence of the oscillations being very stable. In comparison, the limit cycles from the opposite chirality scenario (figure 28b) are not as circular and they are noisier. The fact that they are not circular means that the phase difference between the

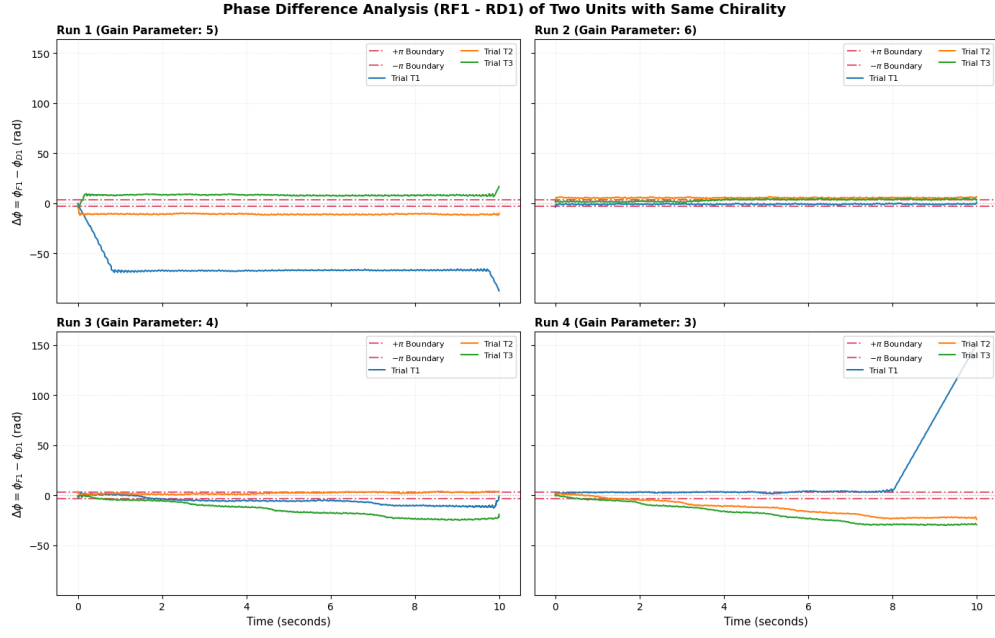
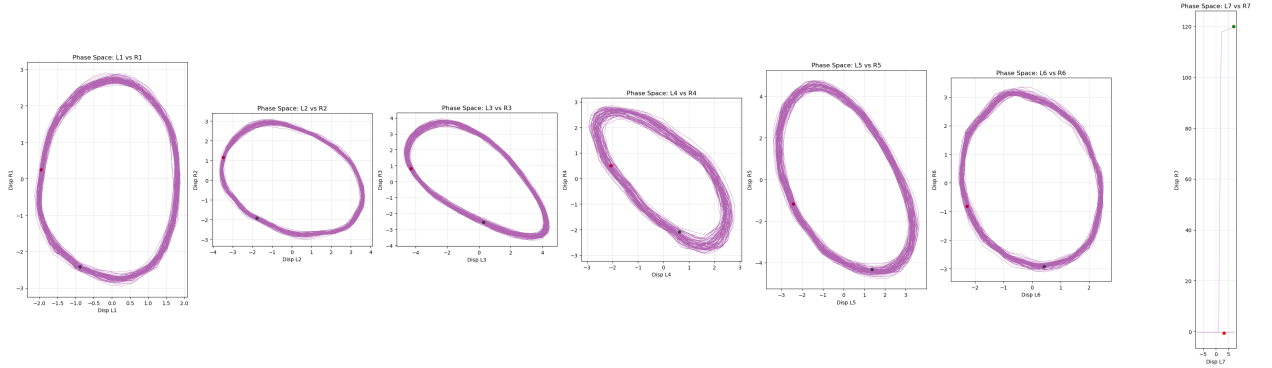
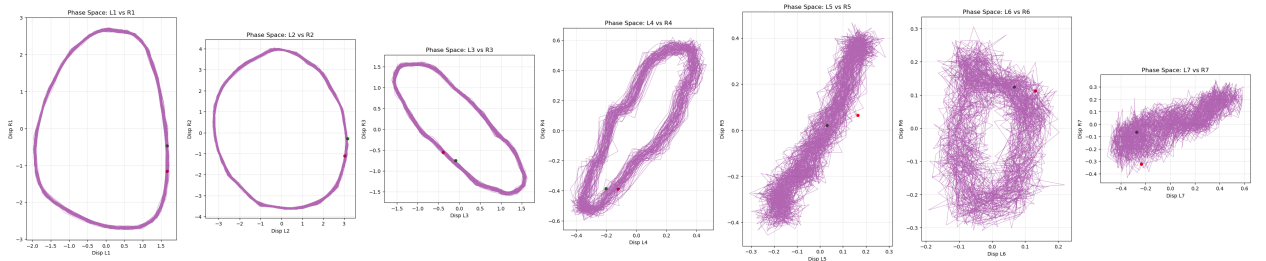


Figure 27: Phase Difference Analysis of Two Units with Same Chirality

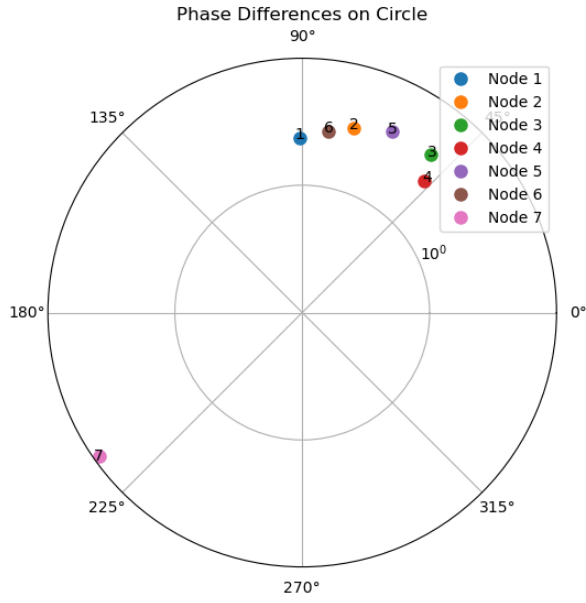


(a) Limit cycles of nodes with same chirality in 1d strip. Results are from measurements done 17-3-2026 Try 1.

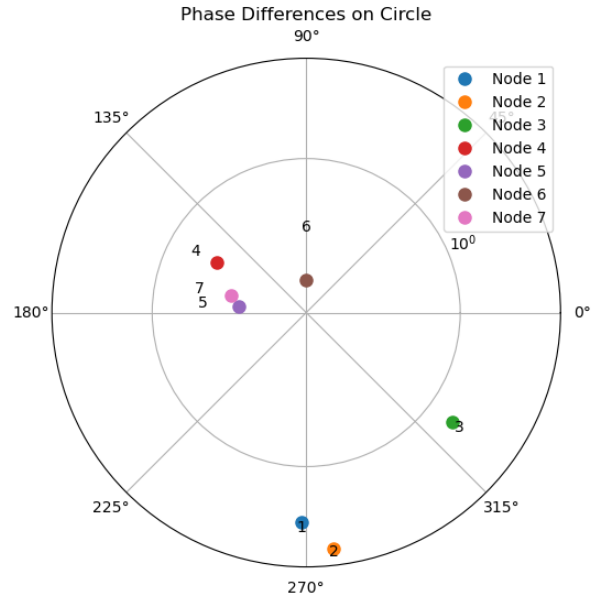


(b) Limit cycles of nodes with opposite chirality in 1d strip. Results are from measurements done 13-3-2026 Try 1.

Figure 28: Comparison of limit cycles for same and opposite chirality scenarios in the 1D strip.



(a) Polar plot of phase difference for actuators within the same unit in scenario with same chirality. The results are from 17-3-2026 Try 1.



(b) Polar plot of phase difference for actuators within the same unit in scenario with opposite chirality. The results are from 13-3-2026 Try 1.

Figure 29: Comparison phase difference of limit cycles for same and opposite chirality scenarios in the 1D strip. The angle in the polar plot corresponds to the phase difference between the two actuators in a unit, and the radius corresponds to the amplitude of the oscillations. More precisely, the radius is the average of the amplitudes of the two actuators in a unit. The radius axis is in logarithmic scale to better visualize small amplitudes.

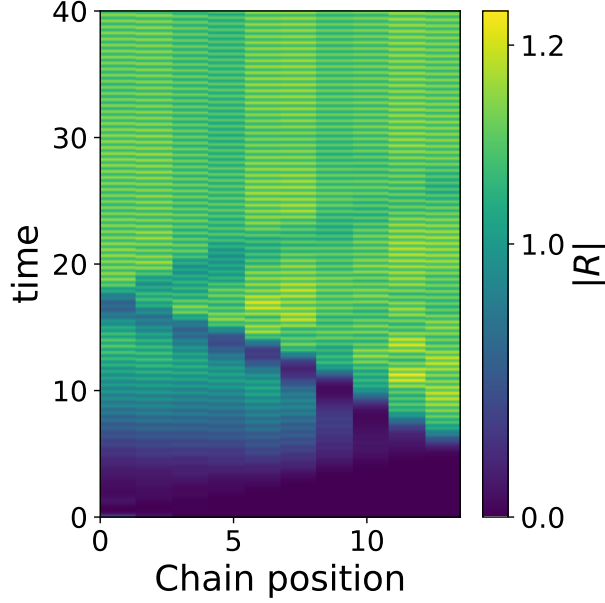


Figure 30: **Kink in amplitude of 1D chain.**

two actuators is not 90° . And they are noisier because their oscillation is weaker, and therefore more affected by neighbours. And the amplitudes are smaller because the nodes don't synchronize.

As the actuators within a unit can deviate from the non-reciprocal behaviour of 90° , we can measure what the phase difference is precisely for each unit. Figures 29 are polar plots of the phase difference. For active units with clockwise/negative chirality the phase difference is -90° . For active units with counterclockwise/positive chirality the phase difference is $+90^\circ$. In the scenario where all active units have the same chirality, we expect to find those nodes, nodes 2, 4 and 6, in the 90° axis. And that is what we observe in figure 29a for nodes 2 and 6. The fact that node 4 is not close to the 90° axis we think is a consequence of the actuators being crooked. In the scenario of opposite chirality, figure 29b, node 2 has clockwise chirality and nodes 4 and 6 remain with counterclockwise chirality. Thus, we expect to find node 2 in the -90° axis and nodes 4 and 6 in the $+90^\circ$ axis. Note that in this experiment, node 2 had a higher gain than nodes 4 and 6, and we saw that the oscillations of 4 and 6 were suppressed in figure 28b. That is why these nodes are closer to the origin in the polar plot of figure 29b. Nevertheless, we still see that node 6 has a strict phase difference of 90° . The phase difference of active units is as one would expect, except for node 4, and doesn't add new information. But what happens to the passive units and their phase difference?

For very disordered activity and a high value of neighbour coupling, one observes a kink in the transient process before synchronization. It is shown in figure 30. The kink corresponds to a unit whose amplitude is very small. It propagates back and forth through the chain, changing direction when reaching the ends of the chain.

E Dynamical matrix Lieb lattice

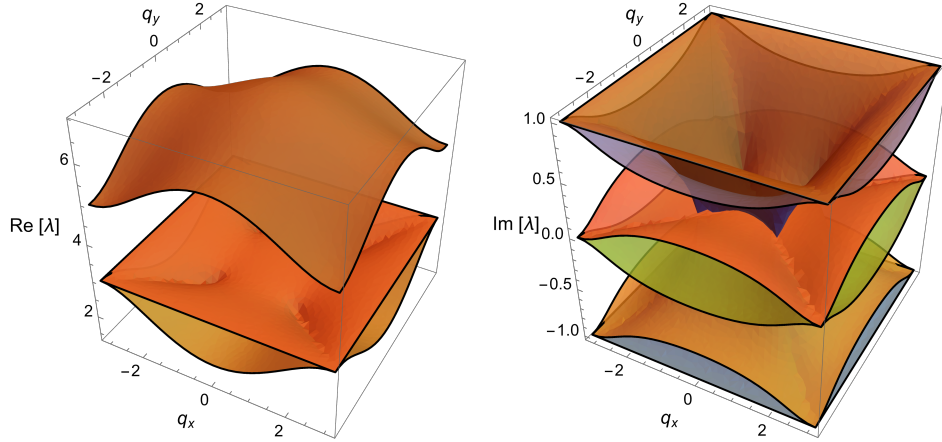


Figure 31: Band structure of bilayer Lieb lattice with non-reciprocal couplings. The eigenvalues are computed from the dynamical matrix in equation (89).

We present the dynamical matrix of a classical Lieb lattice of oscillators with non-reciprocal couplings. The matrix is obtained from Newton's equations of motion, and it describes the linearized dynamics of the system. We are still considering the bilayer Lieb lattice, so there are 6 sites per unit cell and thus the dynamical matrix is a 6×6 matrix. It takes the form in momentum space

$$\mathbf{D} = \begin{pmatrix} 4k_b + k_0 & k_a \sigma_A & -k_b(1 + e^{-iq_x a}) & 0 & -k_b(1 + e^{-iq_y a}) & 0 \\ -k_a \sigma_A & 4k_b + k_0 & 0 & -k_b(1 + e^{-iq_x a}) & 0 & -k_b(1 + e^{-iq_y a}) \\ -k_b(1 + e^{iq_x a}) & 0 & 2k_b + k_0 & k_a \sigma_B & 0 & 0 \\ 0 & -k_b(1 + e^{iq_x a}) & -k_a \sigma_B & 2k_b + k_0 & 0 & 0 \\ -k_b(1 + e^{iq_y a}) & 0 & 0 & 0 & 2k_b + k_0 & k_a \sigma_C \\ 0 & -k_b(1 + e^{iq_y a}) & 0 & 0 & -k_a \sigma_C & 2k_b + k_0 \end{pmatrix}. \quad (89)$$

k_0 denotes the linear bound strength, k_b the coupling strength between nearest neighbours, k_a the non-reciprocal coupling strength, q_x and q_y the momenta in the x and y directions, and σ_A , σ_B , σ_C are the chiralities of the sites A, B and C respectively. We will choose the chiralities as $\sigma_A = 0$, $\sigma_B = -1$ and $\sigma_C = 1$, to reproduce the Lieb lattice. The band structure of the matrix is displayed in figure 31.